



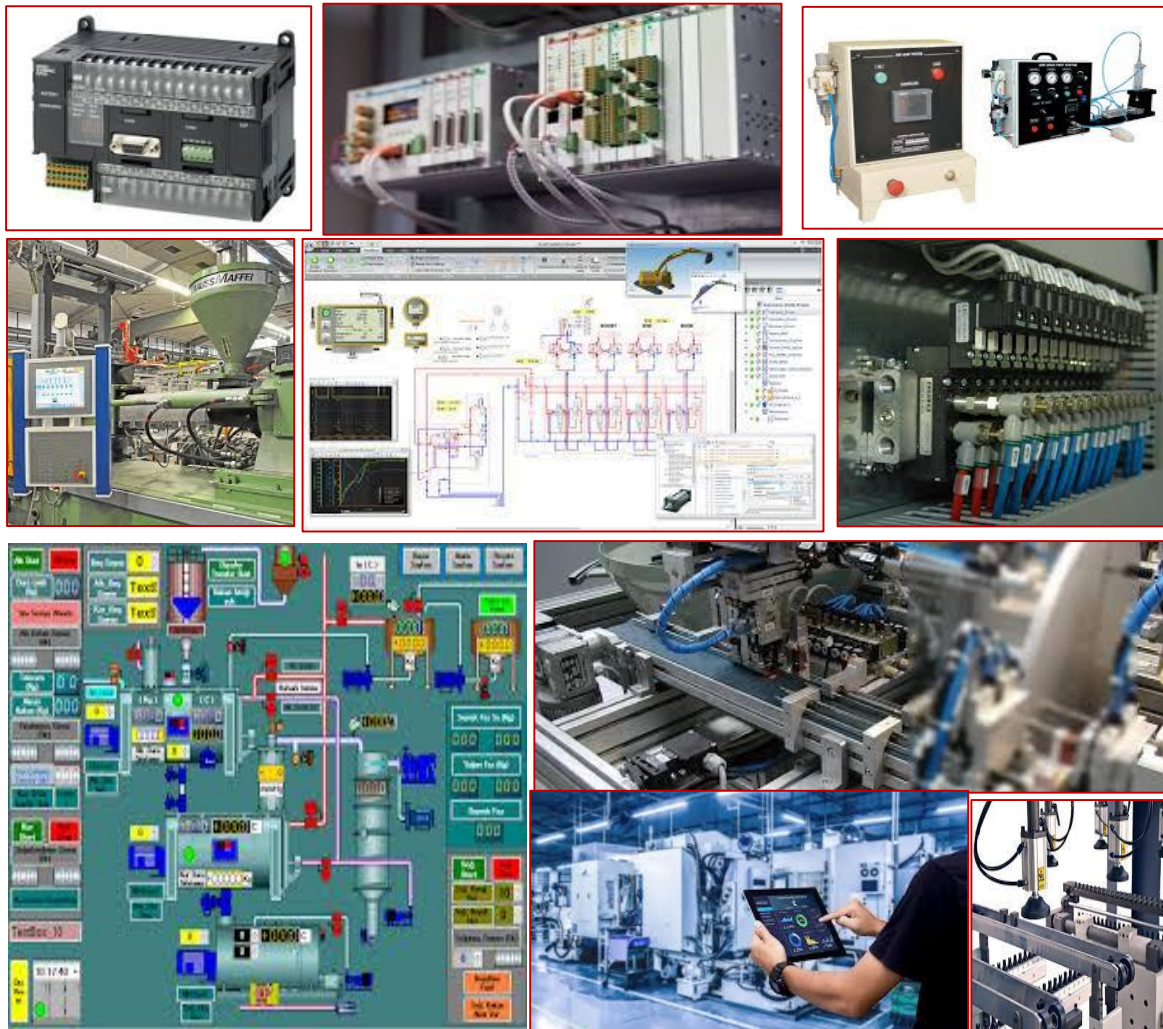
**AMRITA SCHOOL OF ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING
Amritanagar, Coimbatore - 641112**

23MEE384 INDUSTRIAL AUTOMATION LABORATORY (MANUAL / RECORD) (6th semester – Mechanical Engineering)

Name: _____ Roll NO. _____

Semester _____ Academic Year: _____

Branch: _____



Our Mission

To provide value-based technical education and mould the character of the younger generation through a system of wholesome learning so that their earnest endeavour to achieve progress and prosperity in life, is matched by an ardent desire to extend selfless service to the society, one complementing the other.



**AMRITA SCHOOL OF ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING
Amritanagar, Coimbatore - 641112**

Certified that this is the bonafide record of work done by

Mr./ Ms. _____

Roll No. _____ Semester _____

(Branch) _____ in the

_____ Laboratory

of this institute during the academic year _____

Faculty – in – charge

Chairman of the Department

Department Seal

The candidate is examined by me / us

On _____ (date)

Examiner 1

Name:

Examiner2

Name:

Table of contents

Sl. No.	Content of the Record	Page No.
1	Course Objective - Course Outcome and Syllabus with CO-PO Articulation Matrix.	5
2	Laboratory component Evaluation pattern and Rubrics used for evaluation.	6
3	General laboratory Instructions.	7
4	Safety precautions to be followed inside the Laboratory	8
5	List of experiments.	11
6	Hydraulic component description, Hydraulic symbol for circuit design, Pneumatic component description and Pneumatic symbol for circuit design	12 to 21
7	Pneumatic exercises	22 to 45
8	Hydraulic exercises	46 to 61
9	PLC exercises	62 to 106

COURSE SYLLABUS AND CO-PO ARTICULATION MATRIX.

23MEE384 INDUSTRIAL AUTOMATION LABORATORY L-T-P-C: 0-0-3-1

Course Objectives

The course will enable the students to

- Developing design skills in circuit design
- Gaining practical experience in PLC programming

Course Outcomes

After successful completion of the course, Students will be able to:

S.No.	Course Outcomes	Knowledge level [Bloom's Taxonomy]
CO01	Design a hydraulic and electro-hydraulic circuit	Apply
CO02	Design and testing of pneumatic and electro-pneumatic circuit	Apply
CO03	Design and testing of electro-pneumatic circuit	Apply
CO04	Develop a PLC program for the given application	Apply

CO-PO Mapping: [affinity': 3 — high; 2- moderate; 1- slightly]

Com	Program Outcomes [POs]												Program Specific Outcomes [PSOs]*			
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO01	3	2	2	1	2					1		1	1			1
CO02	3	2	2	1	2					1		1	1			1
CO03	3	2	2	1	2					1		1	1			1
CO04	3	2	2	1	2					1		1	1			1

Syllabus

Automation: Introduction to the Fluid power system, components, Symbols, Valves, circuit design, Hydraulic, Pneumatic, Electro-Hydraulic, Electro-pneumatic circuit design, and PLC Programming.

Reference Books:

1. Laboratory Manuals
2. Antony Esposito, "Fluid Power with Applications", Pearson, Seventh Edition, 2010.
3. Petruzella, Frank D. Programmable logic controllers. Fourth Edition, Tata McGraw-Hill Education, 2010

List of Equipment required for meeting the COs

1. Automation Studio V7 Software / FluidSim V6.1
2. Basic Pneumatic Trainer kit
3. Basic Electro-Pneumatic Trainer kit
4. Add on PLC Trainer kit
5. Modular Production System (MPS) with two stations

EVALUATION PATTERN

Assessment	Weightage in %
Continuous Assessment (CA) Before Mid term (6 week task or experiments)	15
Mid term examination	20
Continuous Assessment (CA) After Mid term (6 week task or experiments)	15
Record submission	10
End Semester	40

Rubrics used for Automation Studio and Record:

	Description	Good	Average	Poor
Automation Studio and Record.	Circuit simulation (10 marks) (Individual evaluation)	Good – 10 Marks (Circuit is adequately made and readily visible on the screen and accurately running)	Average – 7 Marks (Circuit is made with few overlapping connections and consumes more space and running)	Poor – 5 Marks (Circuit is incomplete and uses inappropriate components – Not running correctly)
	Development of circuit using the hardware (10 marks)(Group evaluation)	Good – 10 Marks (Circuit has developed appropriate and optimal usage of components and space and accurately running)	Average – 7 Marks (Circuit has developed with more components and consumes more space and running)	Poor – 5 Marks (Circuit is incomplete and not running correctly)
	Record Submission (10 marks) (Individual evaluation)	10 Marks – On-time submission with correct circuit (Next lab class)	7 Marks – Delay in submission with circuit in error (Second next lab class)	5 Mark – vary late submission and wrong circuit

Rubrics used for PLC evaluation

	Program Logic (20 Marks)	Variables Used (10 Marks)	Error identification and debugging (15 Marks)	Application (15 Marks)
PLC	0-6 Marks: Incomplete or incorrect program logic.	0-3 Marks: Limited or incorrect use of variables.	0-5 Marks: Inefficient debugging process.	0-5 Marks: Limited understanding of control system applications.
	7-13 Marks: Adequate program logic with some errors.	4-7 Marks: Adequate use of variables with some inaccuracies.	6-10 Marks: Adequate debugging with some inefficiencies.	6-10 Marks: The application is adequate, with some errors.
	14-20 Marks: Well-structured and error-free program logic.	8-10 Marks: Proper and efficient utilization of variables.	11-15 Marks: Efficient and systematic debugging process.	11-15 Marks: Effective application of PLC in control systems.

GENERAL LABORATORY INSTRUCTIONS

1. Come to the lab in uniform.
2. Enter the lab with Identity Card.
3. During the lab session, your first objective is to learn about the system, component or controller that you are working on.
4. Lab report should be self-content and neatly written. They should contain your circuit diagrams as well as an explanation.
5. The exercises on the laboratory record are base line exercises. You are encouraged to formulate additional questions and to test them using the lab set up. How much you learn depends on how willing you are to ask additional questions.
6. Expect to do most of your learning in the lab or in front of your computer, rather than from lectures.
7. Your lab report and your modeling exercises should be your own work, and reflect your understanding of the problem. No copying allowed, you are encouraged to work together as a group.
8. Fair records are to be submitted within one week of doing the exercise.
9. Listen to the instructions carefully.
10. Design your circuit before the lab session.
11. Follow the lab manual / record; don't create your own symbols.
12. Non-labeled diagrams are considered wrong diagrams.
13. Don't start constructing your design before you have all the instructions.
14. Unplug the power source (Pneumatic or electrical) while constructing your design.
15. Demonstrate your working design.
16. After simulating the circuit remove the components and put them in place.
17. Lab groups will be composed of minimum three students, so carryout the simulation with the team sprit and group learning.

SAFETY PRECAUTIONS TO BE FOLLOWED INSIDE THE LABORATORY

1. DETAILS OF EQUIPMENT

- FESTO PNEUMATIC TRAINER KIT
- FESTO ELECTRO-PNEUMATIC TRAINER KIT
- PLC - Controllers CECC
 - 12 digital inputs, 8 digital outputs, additionally 2 high-speed counters up to 180 kHz
 - Ethernet 10/100 MBPS, Modbus TCP client/server, EasyIP, TCP/IP, OPC server available
 - 2 RS232 interfaces
 - 1 RS422/RS485 interface allows freely configurable communication with different devices
- MPS
 - Operating Pressure: 6 bar
 - Operating voltage: 24 V DC, 4.5
 - Digital inputs/outputs: 6/4

General safety

- Trainees may only work with the circuits under the supervision of a trainer.
- Observe the specifications included in the technical data for the individual components, particularly all safety instructions.
- Malfunctions that may impair safety must not be generated in the training environment and must be eliminated immediately.
- Wear personal protective equipment (safety goggles, safety shoes) when working on circuits.

Mechanical safety

- Switch off the power supply– Switch off the working and control power before working on the circuit.
- Only reach into the setup when it's at a complete standstill.
- Observe possible overrunning of the drives.
- Mount all of the components securely onto the slotted profile plate.
- Make sure that limit switches are not actuated from the front.
- Risk of injury during troubleshooting
 - Use a tool to actuate the limit switches, for example, a screwdriver.
 - Set all components up so that activation of switches and disconnectors is not made difficult.

- Follow the instructions regarding the positioning of the component

Electrical safety

- Disconnect from all sources of electrical power
- Switch off the power supply before working on the circuit.
- Use extra-low voltage only: max. 24 V DC.
- Establishing and disconnecting electrical connections
 - Electrical connections may only be established in the absence of voltage.
 - Electrical connections may only be disconnected in the absence of voltage.
- Maximum permissible current loads for cables and devices must not be exceeded.
 - Always compare the current ratings of the device, the cable, and the fuse.
 - If these differ, use a separate upstream fuse for appropriate overcurrent protection.
- Use only connecting cables with safety plugs for electrical connections.
- When laying connecting cables, make sure they are not kinked or pinched.
- Do not lay cables over hot surfaces.
- Hot surfaces are identified with a corresponding warning symbol.
- Make sure that connecting cables are not subjected to continuous tensile loads.
- Devices with an earth terminal must always be grounded.
- The device is not equipped with an integrated fuse unless specified otherwise in the technical data.
- Always pull on the plug when disconnecting connecting cables; never pull the cab.

Pneumatic safety

- Depressurise the system
- Switch off the compressed air supply before working on the circuit.
- Check the system with pressure measuring instruments to ensure the entire circuit is pressure-free.
- Do not exceed the maximum permissible pressure of 600 kPa (6 bar).
- Do not switch on the compressed air until all tubing connections have been completed and secured.
- Do not disconnect the tubing while under pressure.
- Do not attempt to connect tubing or push-in connectors with your hands or fingers.
- Risk of injury when switching compressed air on.
- Cylinders may advance and retract automatically.
- Risk of accident due to advancing cylinders.
 - Always position pneumatic cylinders so that the piston's working space is unobstructed.

- Ensure that the piston rod cannot collide with any rigid setup components.
- Risk of accident due to tubing slipping off - Use the shortest possible tubing connections.
- If tubing slips off - Switch off the compressed air supply immediately.
- Pneumatic circuit setup:
 - Connect the devices with plastic tubing with an outside diameter of 4 or 6 mm. Push the tubing into the push-in connector as far as it will go.
 - Switch off the compressed air supply before dismantling the circuit.
- Dismantling the pneumatic circuit
 - Press the blue release ring down so the tubing can be pulled out.
- Noise due to escaping compressed air
 - Noise caused by escaping compressed air may damage your hearing.
 - Reduce noise by using silencers or wearing hearing protection if noise cannot be avoided.
 - All of the exhaust ports of the components included in the equipment set are equipped with silencers.
 - Do not remove the silencers.

LIST OF EXPERIMENTS & MARKS

Ex. No.	List of experiments	Simulation (Out of 10)	Hardware (Out of 10)	Signature of the faculty with date.
1	Design, simulation and testing of pneumatic circuits using single acting and double acting cylinder			
2	Design, simulation and testing of pneumatic circuit using logic elements, flow control valves and quick exhaust valves.			
3	Design, simulation and testing of multiple actuator pneumatic circuits using cascade method for various industrial applications.			
4	Design and simulation of basic electro-Pneumatic circuits.			
5	Design, simulation and testing of electro-pneumatic circuits with multiple actuator with and without relay for various industrial applications.			
6	Design and simulation of hydraulic circuits circuit using single acting and double acting cylinder			
7	Design and simulation of hydraulic circuits using sequence valve and counterbalance valve applications.			
8	Design and simulation of basic electro-hydraulic circuits			
9	Design and simulation of multiple actuator hydraulic and electro-hydraulic circuits for various industrial application with relay and without relay.			
10	Ladder programming for Boolean operations and math operations			
11	Interfacing of electro-pneumatic system with PLC			
12	Interfacing of electro-pneumatic system with PLC, timer, and counter			
13	Design of mechatronics system using distribution station with PLC			
14	Design of mechatronics system using sorting system with PLC			
15	Designing the setup for a factory and programming the PLC with various applications			

INTRODUCTION TO FLUID POWER SYSTEMS

FLUID POWER

- Energy is transmitted and controlled through pressurized fluid, either liquid or gas, in fluid power. The term fluid power applies to both hydraulics and pneumatics.
- Hydraulics uses pressurized liquid, for example, oil or water;
- Pneumatics uses compressed air or other neutral gases.
- Fluid power can be effectively combined with other technologies through sensors, transducers, and microprocessors.

Machines often need means to transmit power over a distance. Power is the rate at which *work* is being done. If you lift a weight, your body is doing work to move the weight up. The work is being done against the force of gravity that is trying to pull the weight down. The faster you lift the weight, the more power you are generating.

There are purely mechanical ways to do this. Think about the drive train of a car, where the torque of the engine is transmitted over a distance, through the gearbox and the drive shaft to the wheels to make the car move. There are also electrical ways to do this, like the power that flows over power lines to make your household appliances work.

Hydraulics and pneumatics are a very common way to transmit power. Systems using hydraulics and pneumatics can be efficient and portable, can produce a lot of power but don't need to weigh very much overall, and are fairly easy to maintain and design. Hydraulics and pneumatics offer efficiency, portability, good power-to-weight ratio and ease of maintenance and design. Hydraulics use liquids, pneumatics uses gases.

Hydraulics use liquids (oil, generally) and pneumatics use gases (air, usually) to transmit power. Some of the systems using hydraulics and pneumatics include

- ❖ Excavators
- ❖ Presses
- ❖ Vehicle Brakes
- ❖ Construction Equipment
- ❖ Automated machinery (grippers, ejectors, actuators)

Advantages of Fluid Power Systems

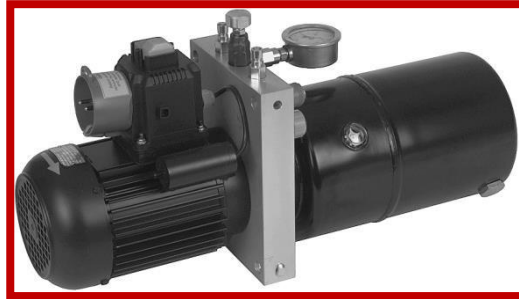
- Multiplication and variation of force
- Easy, accurate control
- Multi-function control
- High horsepower, low weight
- Low speed torque
- Constant force or torque
- Safety in hazardous environments
- Established standards and engineering

HYDRAULIC SYSTEM

Components of power supply section of hydraulic system

The most important components in this section are:

- ❖ Drive
- ❖ Pump
- ❖ Pressure relief valve
- ❖ Coupling
- ❖ Reservoir
- ❖ Filter
- ❖ Cooler
- ❖ Heater



In addition, every hydraulic system contains service, monitoring and safety devices and lines for the connection of hydraulic components. Hydraulic systems (with the exception of hand pumps) are driven by motors (electric motors, combustion engines). Electrical motors generally provide the mechanical power for the pump in stationary hydraulics, whilst in mobile hydraulics combustion engines are normally used. In larger machines and systems, the central hydraulics is of importance.

All consuming devices in a system with one or several hydraulic power supply units and with the help of one or more reservoirs are supplied via a common pressure line. The hydraulic reservoir stores hydraulic power which is released as required. Pressure, return and waste oil lines are all ring lines. Space and power requirements are reduced by employing this type of design.

The pump in a hydraulic system, also known as a hydraulic pump, converts the mechanical energy in a drive unit into hydraulic energy (pressure energy). The pump draws in the hydraulic fluid and drives it out into a system of lines. The resistances encountered by the flowing hydraulic fluid cause a pressure to build up in the hydraulic system.

The level of the pressure corresponds to the total resistance which results from the internal and external resistances and the flow rate.

- External resistances: come about as a result of maximum loads and mechanical friction and static load and acceleration forces.
- Internal resistances: come about as a result of the total friction in the lines and components, the viscous friction and the flow losses (throttle points).

Thus, the fluid pressure in a hydraulic system is not predetermined by the pump. It builds up in accordance with the resistances – in extreme cases until a component is damaged. In practice, however, this is prevented by installing a pressure relief valve directly after the pump or in the pump housing at which the maximum operating pressure recommended for the pump is set.

- ❖ Convert power from rotating shafts into fluid flow

- ❖ Positive displacement devices generate flow, not pressure, blocking of the pump output creates pressure

Types of Pumps

Rarely found in power hydraulic systems

- ❖ Turbine
- ❖ Impeller
- ❖ Propeller

Common in power hydraulic systems

- ❖ Gear
- ❖ Vane
- ❖ Piston
- ❖ Gerotor

Piston pumps are preferred for high pressures (2500 – 4000 psi)

Gear pumps are better today than 20 years ago (now used up to 3000 psi)

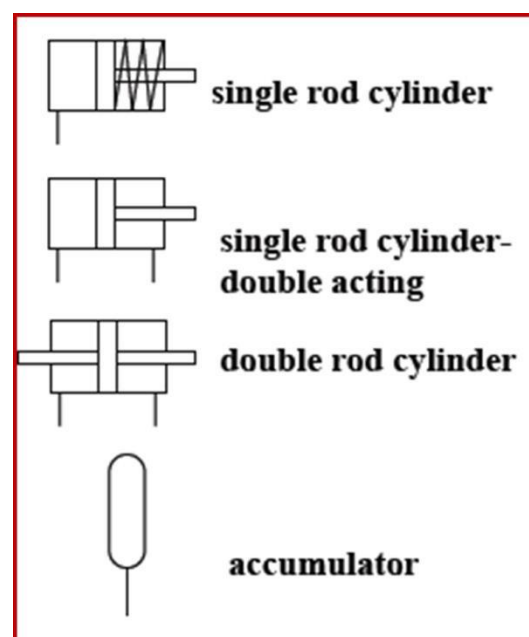
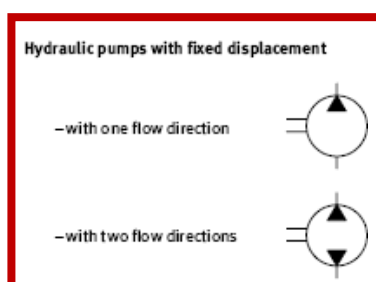
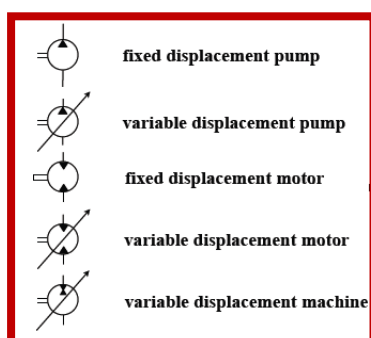
Two classes: 1) fixed delivery and 2) variable delivery pumps.

Limitations of Hydraulic Systems

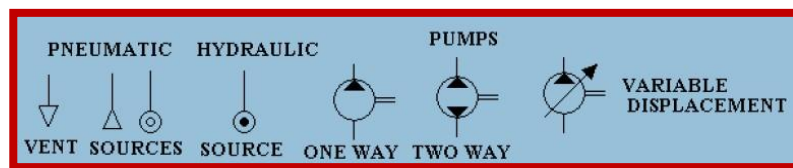
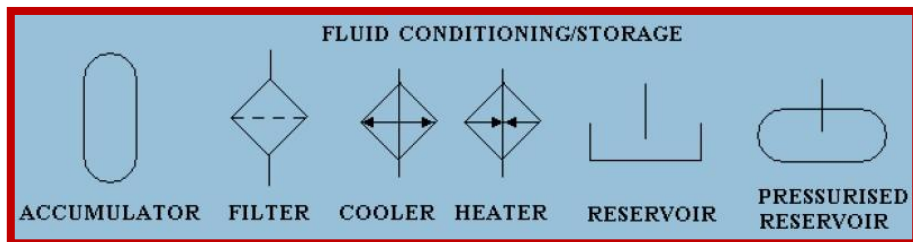
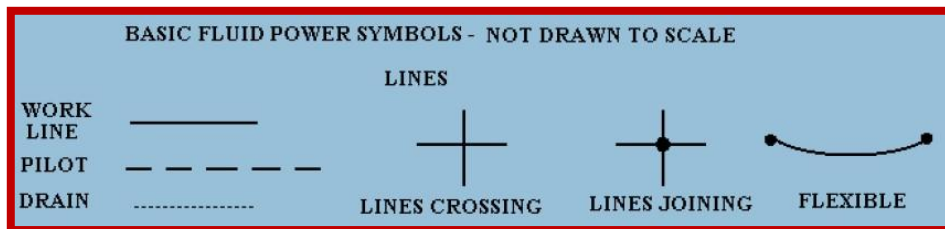
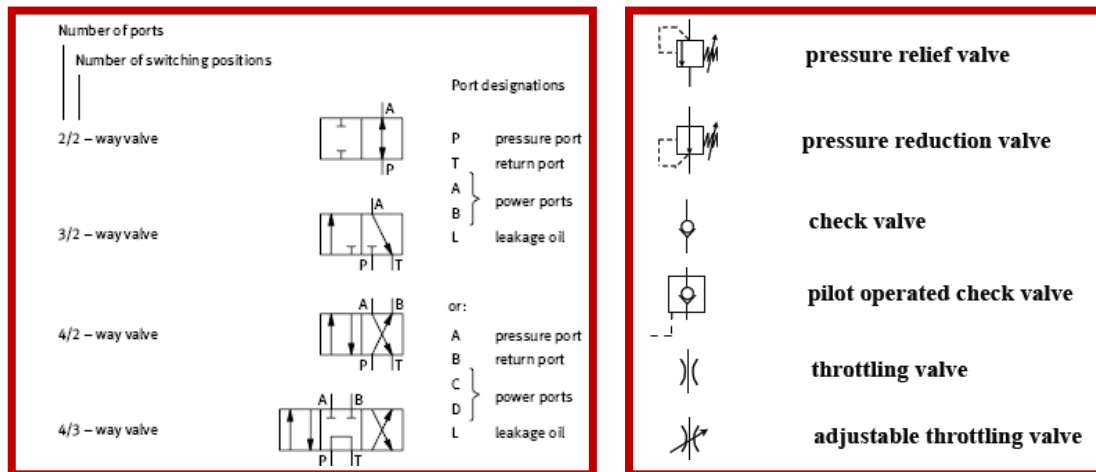
- Hydraulic systems can leak oil at connections and seals.
- Hydraulic power is not as easy to generate as electric power and requires a heavy, noisy pump.
- Hydraulic fluids can cavitate and retain air resulting in spongy performance and loss of precision.
- Hydraulic and pneumatic systems become contaminated with particles and require careful filtering.
- The physics of fluid power is more complex than that of electric motors which makes modeling and control more challenging.

ISO Symbols for Circuit Design

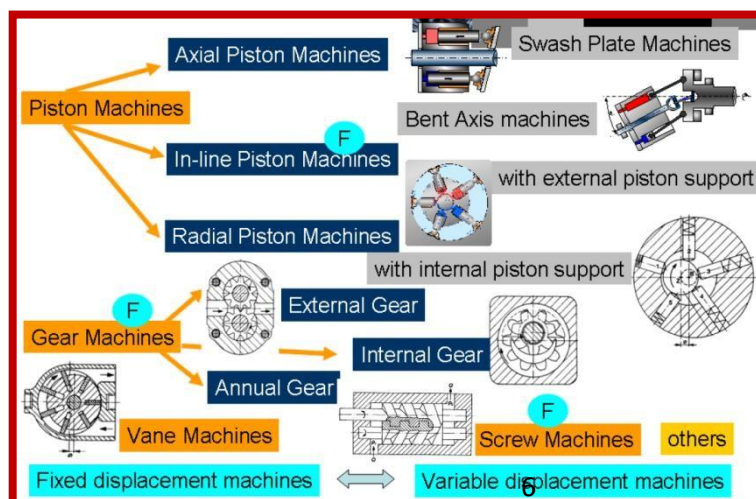
MOTORS AND PUMPS ACTUATORS AND ACCUMULATORS



VALVE SYMBOLS



DISPLACEMENT PUMPS



CLASSIFICATION OF PUMPS

- All pumps may be classified as either positive-displacement or non-positive-displacement. Most pumps used in hydraulic systems are positive-displacement.

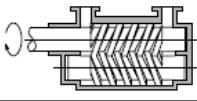
NON-POSITIVE –DISPLACEMENT PUMPS:

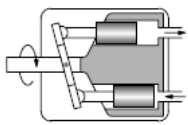
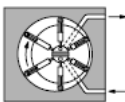
- A **non-positive-displacement pump** produces a continuous flow.
- However, because it does not provide a positive internal seal against slippage, its output varies considerably as pressure varies.
- Centrifugal and propeller pumps are examples of non-positive-displacement pumps.
- If the output port of a non-positive-displacement pump were blocked off, the pressure would rise, and output would decrease to zero.
- Although the pumping element would continue moving, flow would stop because of slippage inside the pump.

POSITIVE –DISPLACEMENT PUMPS:

- In a positive-displacement pump, slippage is negligible compared to the pump's volumetric output flow. If the output port were plugged, pressure would increase instantaneously to the point that the pump's pumping element or its case would fail (probably explode, if the drive shaft did not break first), or the pump's prime mover would stall.
- A **positive-displacement pump** is one that displaces (delivers) the same amount of liquid for each rotating cycle of the pumping element. Constant delivery during each cycle is possible because of the close-tolerance fit between the pumping element and the pump case. That is, the amount of liquid that slips past the pumping element in a positive-displacement pump is minimal and negligible compared to the theoretical maximum possible delivery.
- The delivery per cycle remains almost constant, regardless of changes in pressure against which the pump is working. Note that if fluid slippage is substantial, the pump is not operating properly and should be repaired or replaced. Positive-displacement pumps can be of either fixed or variable displacement. The output of a fixed displacement pump remains constant during each pumping cycle and at a given pump speed. The output of a variable displacement pump can be changed by altering the geometry of the displacement chamber.
- Other names to describe these pumps are hydrostatic for positive-displacement and hydrodynamic pumps for non-positive-displacement. Hydrostatic means that the pump converts mechanical energy to hydraulic energy with comparatively small quantity and velocity of liquid. In a hydrodynamic pump, liquid velocity and movement are large; output pressure actually depends on the velocity at which the liquid is made to flow.

CHARACTERISTIC VALUES FOR THE MOST COMMON CONSTANT PUMPS

	Types of design	Speed range r.p.m.	Displacement volume (cm ³)	Nominal pressure (bar)	Total efficiency
	Gear pump, internally toothed	500 – 3500	1.2 – 250	63 – 160	0.8 – 0.91
	Gear pump, externally toothed	500 – 3500	4 – 250	160 – 250	0.8 – 0.91
	Screw pump	500 – 4000	4 – 630	25 – 160	0.7 – 0.84

	Rotary vane pump	960 – 3000	5 – 160	100 – 160	0.8 – 0.93
	Axial piston pump	 – 3000 750 – 3000 750 – 3000	100 25 – 800 25 – 800	200 160 – 250 160 – 320	0.8 – 0.92 0.82 – 0.92 0.8 – 0.92
	Radial piston pump	960 – 3000	5 – 160	160 – 320	0.90

HYDRAULIC ACTUATORS

- A hydraulic actuator receives pressure energy and converts it to mechanical force and motion. An actuator can be linear or rotary. A linear actuator gives force and motion outputs in a straight line. It is more commonly called a cylinder but is also referred to as a ram, reciprocating motor, or linear motor. A rotary actuator produces torque and rotating motion. It is more commonly called a hydraulic motor or motor.
- There are two basic types of hydraulic cylinder
 - Single-acting and
 - Double-acting cylinders.

In a differential cylinder, the areas where pressure is applied on a piston are not equal. On a head end, a full piston area is available for applying pressure. At a rod end, only an annular area is available for applying pressure. A rod's area is not a factor, and what space it does take up reduces the volume of oil it will hold.

Two general rules about a differential cylinder are that-

- With an equal GPM delivery to either end, a cylinder will move faster when retracting because of a reduced volume capacity.

- With equal pressure at either end, a cylinder can exert more force when extending because of the greater piston area. In fact, if equal pressure is applied to both ports at the same time, a cylinder will extend because of a higher resulting force on a head end.

HYDRAULIC MOTORS

- Hydraulic motors convert hydraulic energy into mechanical energy. In industrial hydraulic circuits, pumps and motors are normally combined with a proper valving and piping to form a hydraulic-powered transmission. A pump, which is mechanically linked to a prime mover, draws fluid from a reservoir and forces it to a motor. A motor, which is mechanically linked to the workload, is actuated by this flow so that motion or torque, or both, are conveyed to the work.
- The principal ratings of a motor are torque, pressure, and displacement. Torque and pressure ratings indicate how much load a motor can handle. Displacement indicates how much flow is required for a specified drive speed and is expressed in cubic inches per revolutions, the same as pump displacement. Displacement is the amount of oil that must be pumped into a motor to turn it one revolution. Most motors are fixed-displacement; however, variable-displacement piston motors are in use, mainly in hydrostatic drives. The main types of motors are gear, vane, and piston. They can be unidirectional or reversible. (Most motors designed for mobile equipment are reversible.)

HYDRAULIC VALVES

- Control valves allow hydraulic systems to produce the type of motion or level of force needed to complete the functions expected of a hydraulic circuit. A variety of valves can control actuator direction, speed, and force output.

The three basic types of control valves are:

- Pressure control
- Directional control
- Flow control

Pressure control valves can:

- Protect the system from damage due to excessive pressure
- Sequence motion
- Limit pressure in selected sections of a circuit
- Directional control valves direct fluid flow to establish and control actuator movement.
- Flow control valves control the operating speed of actuators
- They provide a means to vary the rate of fluid flow

Pressure control valves may be grouped into one of five types

- System maximum pressure control
- Actuator sequence control
- Restrained movement control
- Pump unloading control

- Reduced pressure control

Control valves can be classified by internal modes of operation

- Direct operation
- Balanced-piston operation
- Direct-operated valves depend on heavy internal springs to establish valve operating pressure
- Balancing-piston valves (Compound relief valves) use lighter springs and system pressure acting on internal valve mechanisms to establish the desired operation
- Maximum system pressure control devices are referred to as:
 - Relief valves
 - Safety Valves
 - Hydraulic pressure fuses

COMPOUND RELIEF PRESSURE CONTROL VALVE

- The pilot section of the compound relief valve contains a small, direct-operated relief valve
- The pilot section indirectly establishes maximum system pressure by controlling the pressure in the balancing-piston section of the valve
- The balancing-piston section of the compound relief valve uses a metering orifice and a balancing spring to create pressure and force differences
- These differences correctly position the piston to produce a desired maximum system operating pressure.

Safety valves are used to prevent damage to the hydraulic system if the system relief valve should fail to open

- Typically, safety valves are direct-operated relief valves
- Safety valves are generally set 25% higher than the normal system operating pressure

Hydraulic pressure fuses function as a pressure-limiting device by using a disk that ruptures at a predetermined pressure

- They act as a positive-pressure-limiting device for systems where system pressure limits are critical to safe system operation

Sequence valves allow the automatic sequencing of two or more actuators in a hydraulic circuit

- Primary actuator moves as soon as fluid flow is directed to the actuator section of the circuit
- Sequence valve blocks flow to the secondary actuator until a predetermined pressure is reached, then allows fluid flow to the actuator
- A sequence valve is typically fitted with an integral check valve
- This allows free flow of fluid around the valve when the direction of the actuator is reversed

COUNTER BALANCE VALVES & BREAK VALVES

- Restrained movement control valves are used in circuits to prevent unexpected actuator movement. Often called:
 - Counterbalance valves when used with cylinders
 - Break valves when used with motors
- Counterbalance and brake valves are normally closed valves
- A check valve is required to permit free reverse flow around to valve
- Counterbalance valve with integral check valve
- Counterbalance valves allow the downward movement of loads supported by a cylinder only when the system pump is operating
- Force generated by system pressure and the force created by the weight of the load are needed to move the load downward
- Brake valves are used to prevent an overrunning load from continuing to turn a hydraulic motor after the directional control valve has been closed
- Overrunning loads can turn the motor into a pump, allowing the motor to turn past selected point

PUMP UNLOADING VALVE

- Pump unloading controls hold a desired operating pressure while the pump operates at near-zero pressure
- This reduces energy consumption and maintenance costs
- Pump unloading controls use an unloading valve and a dual pump or accumulator to maintain desired system pressure while dumping unneeded pump output to the reservoir at very low pressure
- Unloading valves are normally closed valves with external pilots
 - Unloading relief valves are used with accumulator circuits
 - Direct-operated valves can be used in typical high-low dual-pump circuits
- Unloading relief valve uses a differential-area plunger mechanism to allow the valve to remain open while unloading the pump to the reservoir
 - Differential area of plunger mechanism is often 85%
 - Once the valve is open, system pressure must drop 15% before the valve will reset

PRESSURE REDUCING VALVE

- Reduced pressure control allows a portion of a system to operate at a pressure below the maximum pressure setting of the system relief valve
 - Allows a system to operate using two or more pressures
 - Maximum pressure is determined by the system relief valve with the additional pressures lower
- Pressure reducing valves may be:
 - Direct operated
 - Balancing piston
- Pressure-reducing valves are normally open valves with an internal pilot and external drain
 - When the valve is operating at the reduced pressure, there is a continual fluid flow through the drain line
 - Any resistance to fluid flow from the drain line reduces valve operating efficiency

SHUT OFF VALVES

The primary purpose of shut-off valves is to block fluid flow through a hydraulic system line

- Globe valve
- Gate valve
- Ball valve
- Spool valve
- Needle valve

CHECK VALVES

- The primary purpose of check valves is to allow free flow in one direction while preventing reverse flow
- Other functions include:
 - Bypassing components during the return cycle of the system
 - Providing flow resistance to maintain a minimum system pressure required for pilot operations
- A standard check valve consists of a valve body containing a one-way valve located between inlet and outlet ports
- The one-way valve allows fluid flow through the valve in only one direction
- Some designs contain a spring that seats the valve poppet or ball
- In other designs, the poppet is seated only by fluid flow
- Restriction check valves allow free flow in one direction and restricted flow when flow direction is reversed
- This is accomplished via a metering orifice machined into the poppet.
- Pilot-operated check valves can allow reverse flow through the valve
 - Typically, pilot pressure opens the valve
 - In some designs, pilot pressure may also hold the valve shut to block flow in both directions

THREE WAY DIRECTION CONTROL VALVES

- Three-way directional control valves provide a means to extend rams and single- acting cylinders
- The actuator is returned to its original position by an external force
 - System load
 - Spring built into the actuator
- During extension, the three-way valve connects the actuator inlet line to a system supply line, allowing fluid to enter and extend the unit
- During retraction, the valve blocks the supply line and connects the actuator line to a system return line, allowing external force to return the actuator to its original position while directing displaced fluid to the reservoir.
- Four-way directional control valves are typically manufactured as two- or three- position valves
- This provides several operating options when designing circuits

- Four-way directional control valves are typically manufactured as two- or three-position valves
- This provides several operating options when designing circuits

ELECTRICAL CONTROL OF HYDRAULIC VALVES

- Electrical control of hydraulic systems is common in many types of equipment
 - Simple solenoid devices to shift basic valves
 - Electronic controllers operate proportional solenoid valves to produce extreme accuracy and repeatability

FLOW CONTROL DEVICES

- Flow control devices produce the desired rate of actuator operating speed by controlling the volume of fluid allowed to reach the actuator
- Flow control devices can be divided into two general types:
 - Restrictor
 - Bypass
- Restrictor-type flow control valves limit the volume of fluid through the valve
- Excess pump output is forced to return to the reservoir through the system relief valve
- The flow rate through a simple, sharp-edged orifice depends on:
 - Area of the orifice
 - Pressure difference between the inlet and outlet sides of the orifice
 - Viscosity of the fluid, which varies with fluid temperature
- Flow control valves may be non compensated or compensated
 - The flow rate through non compensated valves varies as the load or fluid viscosity changes
 - Compensated valves automatically adjust for fluid pressure variations to produce a consistent flow rate under varying load and temperature conditions
- Non compensated and compensated flow control valves may have:
 - Fixed flow rate
 - Adjustable flow rate
- The simplest restrictor-type flow control valve is a simple orifice
 - Basically a calibrated hole
 - Serves as a non compensated, fixed-rate flow control device
- A needle valve is the simplest restrictor-type, non compensated adjustable flow control device
 - Consists of an orifice fitted with a tapered needle machined on a threaded stem
 - Turning the threaded stem changes the effective area of the orifice, which adjusts the flow rate through the valve
- When using a restrictor-type, non compensated flow control valve, actuator speed varies when system loads change
- Caused by the change in pressure drop across the control valve, which varies the flow rate through the valve
- A pressure compensator maintains a constant pressure difference across the metering orifice of a flow control valve
 - Senses pressure on the inlet and outlet sides of the orifice

- These pressures generate forces that act on the end surfaces of a sliding spool that is preloaded by a biasing spring
- Force generated by the biasing spring establishes the constant pressure difference across the orifice
- This constant pressure difference maintains constant fluid flow through the valve even when system loads change

VARIOUS HYDRAULIC CIRCUITS

- Safety is always a factor when designing hydraulic circuits
 - High pressures
 - High fluid flow rates
 - Substantial forces
 - Rapid movement of system components
- Inappropriate use of any of these elements can produce a dangerous work situation for a machine operator
- Safety-related circuits ensure:
 - Equipment can be operated without injury
 - Equipment cannot unexpectedly operate
 - Costly equipment is protected against excessive pressure and operating rates

Back-pressure check valve

- A spring-loaded check valve used with some pilot-controlled hydraulic systems to maintain sufficient pressure during the idle portion of the cycle to shift directional control valves.

Bleed-off circuit

- A hydraulic speed-control circuit in which the required flow rate to the actuator is set by returning unneeded pump output directly to the reservoir through a flow control valve.

Deceleration valve

- A control valve specifically designed to allow maximum, deceleration, and feed flow rates. The valves often use mechanical activation and a check valve that allows an unrestricted return flow rate.

High-low pump circuit

- A circuit using two fixed-displacement pumps. The pumps are connected, with appropriate pressure control valves, to provide a high-volume flow at low pressure and a low-volume flow when high pressure is required to operate a system.

Maximum operating pressure

- The highest pressure recommended for operation of a hydraulic or pneumatic system.

Meter-in circuit

- A fluid power flow-control circuit in which fluid is metered into an actuator to control operating speed.

Meter-out circuit

- A fluid power flow-control circuit in which fluid is metered out of an actuator to control operating speed.

Rapid-advance-to-work circuit

- A fluid power circuit design that allows a tool or other machine element to rapidly move to a preset position, slow to a point where machining is to occur, and then move at an accurate cutting speed. May also allow rapid retraction of the work elements.

Regenerative-cylinder-advance circuit

A hydraulic circuit design in which fluid discharged from the rod end of a cylinder combines with pump delivery and is directed to the cylinder blind end. The design results in increased cylinder extension speed, but reduced force capability

Remote location

- A location for control components that is beyond the distance usually used for placement of components on machines.

Sequence

- The order in which various valve and actuator functions occur in a fluid power circuit.

Synchronization

- The timing of one movement or operation with another, such as the identical movement of two actuators operating in a fluid power circuit.

Two-hand safety circuit

- A fluid power circuit in which machine activation and continued operation requires both hands of the machine operator to continuously depress control valves.

PNEUMATIC SYSTEM

Pneumatics has for considerable time been used for carrying out the simplest mechanical tasks, but in recent times has played a more important role in the development of pneumatic technology for automation.

In majority of applications compressed air is used for one or more of the following functions.

- ❖ The use of sensors to determine status of processors
- ❖ Information processing
- ❖ Switching of actuators by final control elements
- ❖ Carrying out the actual work

Pneumatic components can perform the following types of motion:

- ❖ Linear

- ❖ Swivel
- ❖ Rotary

Some industrial applications employing pneumatics are listed below:

General methods of material handling:

- Clamping
- Shifting
- Positioning
- Orienting

General applications:

- ❖ Packaging
- ❖ Feeding
- ❖ Metering
- ❖ Door or Chute control
- ❖ Transfer of materials
- ❖ Turning and Inverting of parts
- ❖ Sorting of parts
- ❖ Stacking of components
- ❖ Stamping and Embossing of components

Pneumatics is used in carrying out machining and working operations. For example:

- ❖ Drilling
- ❖ Turning
- ❖ Milling
- ❖ Sawing
- ❖ Finishing
- ❖ Forming

The product development in Pneumatics can be considered in a number of areas:

- ❖ Actuators
- ❖ Sensors and Input devices
- ❖ Processors
- ❖ Control systems
- ❖ Accessories

Each of the above product groups is important in the development of pneumatic solutions. The demands are for components with,

- ❖ Accessibility for repair and maintenance
- ❖ Low cost of replacement
- ❖ Ease of mounting and connection
- ❖ Low plant maintenance requirements
- ❖ Interchangeability and flexibility

- ❖ Compact design
- ❖ Costs commensurate with the above
- ❖ Readily available
- ❖ Documentation support
- ❖ Minimum training required to support the product

Pneumatics-An introduction:

The surface of the earth is covered entirely by layers of air. It is an abundant gas mixture comprising

- ❖ Nitrogen (approximately 78% volume)
- ❖ Oxygen (approximately 21% volume)

When air is compressed and stored it can be used as a medium for making measurements and for controlling and operating equipment and plant. This is known as pneumatics. The word "Pneuma" comes from Greek and means breath or wind. The word "Pneumatics", the study of air movement and air phenomena is derived from the word "Pneuma".

PRESSURE:

Pressure is force acting at right - angles onto a surface area. One (1) Pascal represents a constant pressure on a surface area of 1 sq. m. with a force of 1N (NEWTON) acting at right - angles to that surface area. 10^5 Pa or 100 K Pa is equal to 1 bar and this represents atmospheric air pressure. As can be seen, the Pascal is a small unit of pressure and hence we tend to use the bar. One (1) bar represents atmospheric pressure.

Pressure is defined as force F per unit area A.

$$P = F/A \quad \text{N/m}^2$$

The unit of pressure is called **Pascal**.

$$1 \text{ Pa} = 1 \text{ N/m}^2$$

$$100,000 \text{ Pa} = 1 \text{ bar}$$

$$100,000 \text{ N/m}^2 = 1 \text{ bar (or)} 1 \text{ bar} = 10 \text{ N/cm}^2$$

ADVANTAGES OF PNEUMATIC SYSTEMS:

- **Availability** - Air is available practically every where in abundant quantities
- **Transport** - Air can be transported in pipelines, even over large distances
- **Storage** - An air compressor need not be in continuous operation. Compressed air can be stored in a reservoir and removed as required.
- **Temperature** - Compressed air is relatively insensitive to temperature fluctuations. This ensures reliable operation, even under extreme conditions.

- **Explosion proof** - Compressed air offers minimal risk of explosion of fire, hence no expensive protection against explosion is required.
- **Cleanliness** - Unlubricated exhaust air is clean. Any lubricated air which escapes through leaking pipes or components does not cause contamination.
- **Components** - The operating components for pneumatics are of simple construction and are therefore relatively inexpensive.
- **Speed** - Compressed air is very fast working medium. This enables high working speeds to be attained.
- **Adjustable** - With compressed air, speeds and forces are variable over wide ranges.
- **Overload safety** - Pneumatic tools and operating components can be loaded to the point of stopping and therefore overload safe.

DISADVANTAGES:

- **Preparation** - Compressed air requires good preparation. Dirt and condensate should not be present.
- **Compressible** - It is not always possible to achieve uniform and constant piston speeds with Compressed air.
- **Force Requirements:** Compressed air is economical only up to a certain force requirement. Under the normal working pressure of 6 to 7 bar and dependent on the travel and speed, the output limit is between 20,000 and 30,000 Newton's.
- Noise Level** - In pneumatic systems exhaust air is loud. This problem has now however, been largely solved due to the development of sound absorption materials and silencers.
- **Costs** - Compressed air is relatively expensive means of converting power. The high energy costs are partially compensated by inexpensive components and higher performance.

BASIC REQUIREMENT FOR MAINS AIR SUPPLIES:

It is essential for proper and effective operation in all pneumatic systems that mains air supplies be

- ❖ Clean
- ❖ Dry
- ❖ Oil free

The preparation of the air starts from the point of generation, Contamination of the air can occur at many potential points in the generation and distribution systems right up to the point of use. The equipment to be considered in the generation, preparation and distribution of air includes:

- ❖ Compressor - It takes air at a lower pressure and delivers it at a higher pressure.
- ❖ Receiver - It is for storing and maintaining constant pressure of the compressed air.
- ❖ Filter - It is for filtering impurities and condensates.
- ❖ Drier - Depending on the application of the compressed air dryers may be used for necessary dehydration of the air.
- ❖ Lubricator - To provide lubrication for sliding parts in Pneumatic devices.
- ❖ Pressure Regulator - To reduce the primary pressure to the most suitable value for the equipment and application.
- ❖ Drainage Point - To drain the water contents in the flow lines and in the tanks.
- ❖ Oil Separator - To separate the compressor oil contents from the compressed air.

ELECTRO-PNEUMATICS:

Integration of pneumatic and electrical technologies has played an important part in the development of a large range of industrial automation solutions. An electro-pneumatic control offers number of advantages to the developer of control systems. Where signals are transmitted over great distances in a control system, the time between transmission and signal reception is minimised using electricity as the transmission medium. Where future growth is anticipated, the incorporation of solenoid actuators and valves provides the option of developing the system to incorporate a controller that allows the system operating cycle to be altered as production requirements change.

An electrical component in electro-pneumatics has the following roles:

- ❖ Sensing Information
- ❖ Processing Information

The sensing of information can be carried out by the roller actuated limit switches, magnetically actuated switches or electronic sensors that sense whether or not a particular operation has occurred. This information is passed to the processing device, which may be a relay. The processor then has the task of directing and modifying its received signal to drive the output stage of the system. This output stage has generally two parts:

First, a final control element that is electrically actuated with a pneumatic output and secondly the pneumatic actuator.

ELECTRO-PNEUMATIC ELEMENTS:

Actuators:

- ❖ Pneumatic cylinders
- ❖ Rotary Actuators
- ❖ Lamps / Buzzers

Control Elements:

- ❖ Solenoid actuated Directional control valves
- ❖ Relays

Processors:

- ❖ Solenoid actuated Directional control valves
- ❖ Logic elements
- ❖ Pneumatic - electric converter
- ❖ Relays

Sensors:

- ❖ Limit switches
- ❖ Push buttons
- ❖ Proximity Sensors

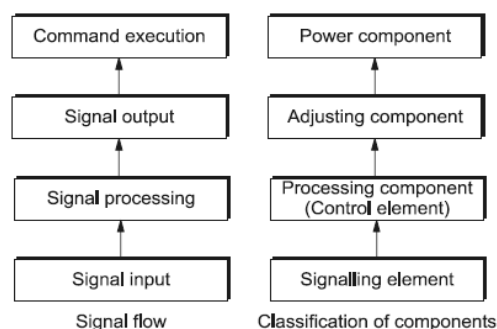
Energy Supply:

- ❖ Compressor
- ❖ Receiver

The succeeding section contains various problems and solutions of pneumatic and electro-pneumatic systems for automation.

SIGNAL FLOW STRUCTURE AND SIGNAL FLOW IN PNEUMATIC DIAGRAM

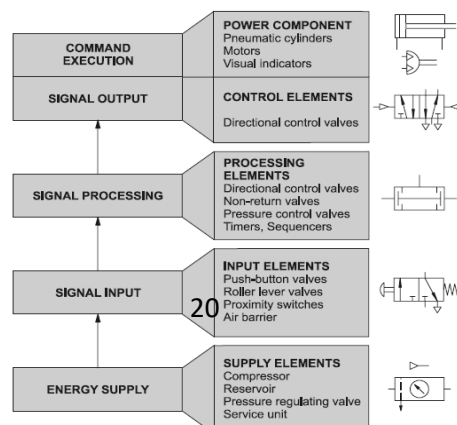
Pneumatic system consists of an interconnection of different groups of elements. This group of elements forms a control path for signal flow, starting from the signal selection (input) through to the actuating section (output). Control elements control the actuating elements in accordance with the signals received from the processing elements.

SIGNAL FLOW

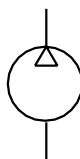
The primary levels in a pneumatic system are :

- ❖ Energy supply, Input elements (sensors)
- ❖ Processing elements (Processors), Control elements
- ❖ Power components (Actuators)

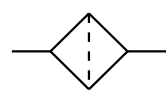
The elements are represented by symbols which indicate the function of the element.

PNEUMATIC CONTROL SYSTEM

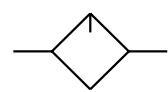
PNEUMATIC COMPONENT SYMBOLS



Air Compressor



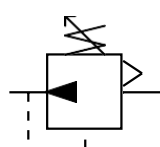
Filter



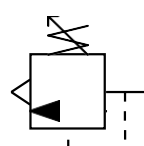
Lubricator



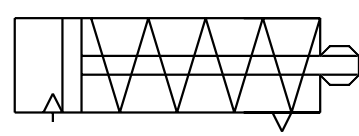
Pressure Gauge



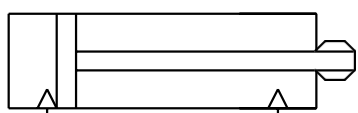
Pressure Regulator



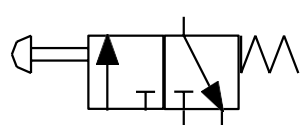
Pressure Relief Valve



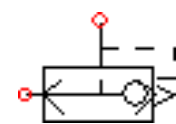
Single Acting Cylinder



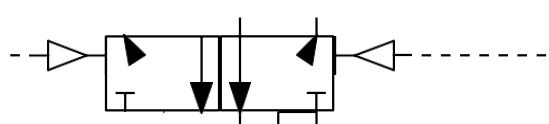
Double Acting Cylinder Valve



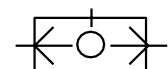
3/2 NC Direction Control Valve
Push button operated spring return



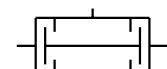
Quick Exhaust



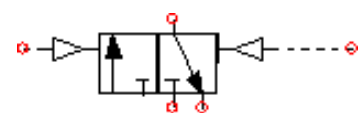
5/2 Direction Control Valve
Double pilot operated



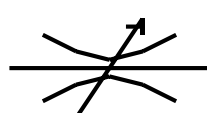
OR valve



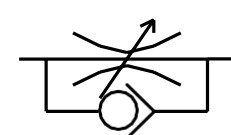
AND valve



3/2 Direction control valve
Double pilot operated



Adjustable flow control valve



One-way flow
(Variable NR Throttle valve)

EXERCISE – 1

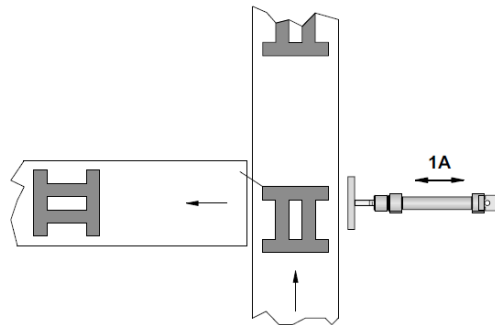
Design, simulation and testing of pneumatic circuits using single acting and double acting cylinder

Exp. No.: 1A**Date:**

Design and simulation of circuits using single acting cylinder for specific applications.

PROBLEM:

Through operation of the push button on the actuating valve, metal stampings lying in random positions are sorted out and transformed to a second conveyor belt. The forward motion of the single acting cylinder is to be adjusted. When the push button is released, the piston rod travels to the retracted end position.

**WORKING PRINCIPLE:**

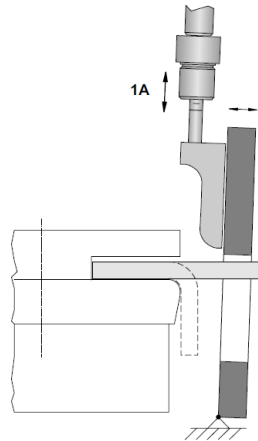
PNEUMATIC CIRCUIT:

INFERENCE:

Exp. No.: 1B	Design and simulation of circuits using double acting cylinder for specific applications.
Date:	

PROBLEM:

Operation of two identical valves by push button causes the forming tool of an edge folding device to thrust downwards and fold over the edge of a flat sheet of cross sectional area 40 x 5. If both or even just one push button is released, double acting cylinder (1A) slowly returns to the initial position.

**WORKING PRINCIPLE:**

PNEUMATIC CIRCUIT

INFERENCE:

EXERCISE – 2

Design, simulation and testing of pneumatic circuit using logic elements, flow control valves and quick exhaust valves.

Exp. No.:2A	Design and simulation of a clamping of workpiece and drilling system.
Date:	

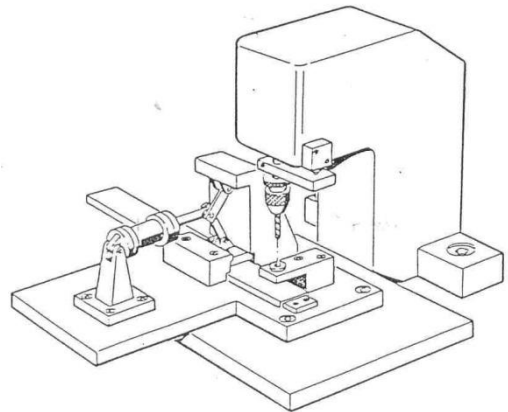
PROBLEM:

Clamping of a work piece:

A flat rectangular plate has to be drilled. Clamping of the plate is done by a double acting cylinder subject to the availability of work piece. The clamping of the plate must be possible by manual control either by pressing a push button or by a foot pedal. After the plate is firmly clamped drilling operation is to be performed. Once the drilling operation is over then only the work piece is unclamped. Unclamping must be initiated by a further manual pushbutton.

Conditions:

- a) Clamping must be possible only when the work piece has been inserted.
- b) Unclamping must not be possible during the drilling operation.
- c) It must be possible to set the clamping operation (advance stroke) to a slow speed.
- d) Unclamping (return stroke) must be carried out very quickly.



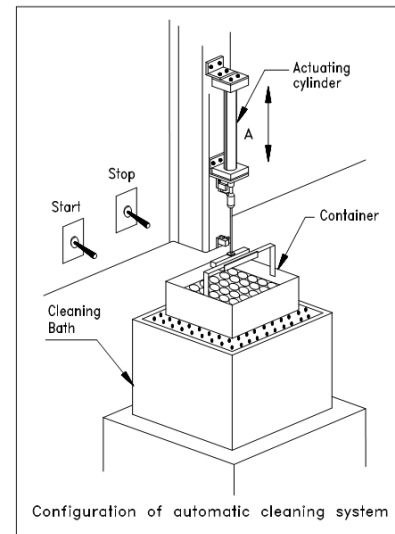
WORKING PRINCIPLE:

PNEUMATIC CIRCUIT:

INFERENCE:

Exp. No.:2B**Date:****Design and simulation of a cleaning bath system.****PROBLEM:**

A container is to be moved up & down in the cleaning bath continuously. The "START" and "STOP" signals for the continuous cycle are to be given by a pushbutton. The effect of the "STOP" signal should be such that the container stops in the raised position. A double acting cylinder is to be controlled such that, after being given the start signal, the piston performs oscillatory motion between the end positions until the reverse signal is input. The cylinder should then remain stationary in the rear end position.

**WORKING PRINCIPLE:**

PNEUMATIC CIRCUIT:

INFERENCE:

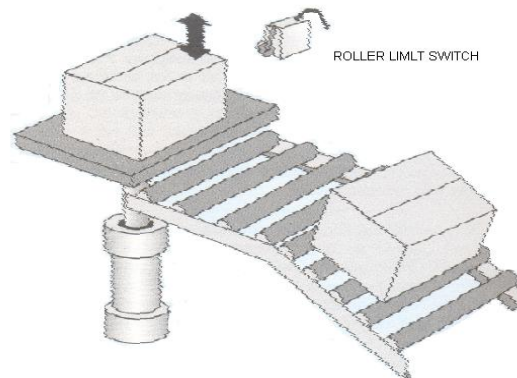
EXERCISE – 3

Design, simulation and testing of multiple actuator pneumatic circuits using cascade method for various industrial applications.

Exp. No.: 3A	Design and simulation of pneumatic lifting system with sensing valve.
Date:	

PROBLEM

Boxes are lifted to conveyor height by a double acting cylinder. The piston rod is to extend fully when a switch is operated. The piston rod is to extended position before the operator can initiate rod retraction by a second switch. A roller limit switch confirms full extension. The rod is to continue forward even if the extension switch is released before full extension is reached.



WORKING PRINCIPLE:

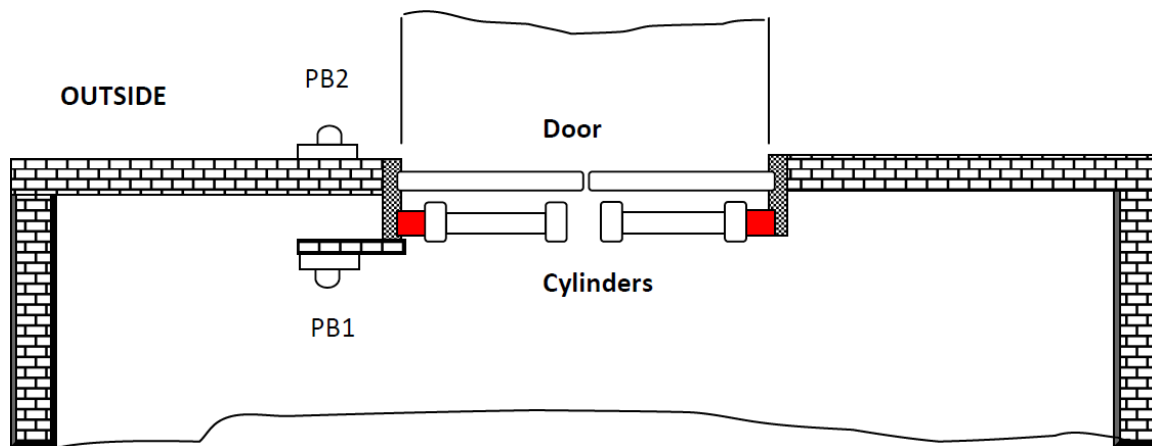
PNEUMATIC CIRCUIT:

INFERENCE:

Exp. No.: 3B	Automatic Open And Door Closing System Using Pneumatic Cylinder
Date:	

PROBLEM:

A pneumatically controlled double sliding door is to be used in an crockery mart. Doublesliding door is to be controlled either from outside or from inside by pressing a pushbutton. The opening and closing operations are initiated by the same pushbutton in each case. The speeds of the double sliding door movements are to be adjustable. Develop a pneumatic control circuit for the given task.



WORKING PRINCIPLE:

PNEUMATIC CIRCUIT:

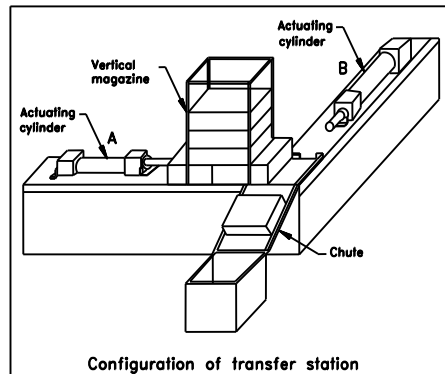
INFERENCE:

Exp. No.: 3C	Pneumatic circuit design for a part transfer station.
Date:	

PROBLEM:

Two cylinders are used to transfer parts from a magazine onto a chute. When a pushbutton is pressed, cylinder (A) extends, pushing the part from the magazine and positions it in preparation for transfer by cylinder (B) onto the out feed chute. Once the part is transferred, the cylinder (A) retracts, followed by the cylinder (B).

Sequence code: A1 B1 A0 B0



WORKING PRINCIPLE:

PNEUMATIC CIRCUIT:

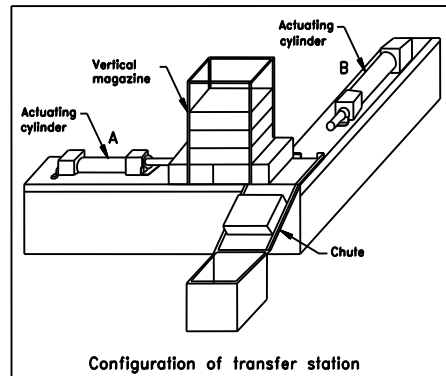
INFERENCE:

Exp. No.: 3D	Pneumatic circuit design for a part transfer station using cascade method.
Date:	

PROBLEM:

Two cylinders are used to transfer parts from a magazine onto a chute. When a pushbutton is pressed, cylinder (A) extends, pushing the part from the magazine and positions it in preparation for transfer by cylinder (B) onto the out feed chute. Once the part is transferred, the cylinder (B) retracts, followed by the cylinder (A).

Sequence code: A1 B1 B0 A0



WORKING PRINCIPLE:

PNEUMATIC CIRCUIT:

INFERENCE:

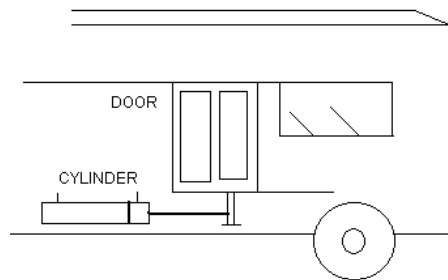
EXERCISE – 4

Design and simulation of basic electro-Pneumatic circuits.

Exp. No.: 4A	Automatic door opening and closing of a bus with passenger sensing.
Date:	

PROBLEM:

The automatic door of a bus is operated by a Bus driver. When the bus driver presses an electrical push button a double acting cylinder extends and opens the door. The passenger entry is monitored by a sensor. This sensor output should be memorized using a relay and its own contact. The driver presses another electrical push button after the confirmation of passenger entry to close the door. The relay should be switched off using the output of door closing limit switch. Both the door opening speed and closing speed is to be adjustable separately. Design and draw an Electro-pneumatic circuit.



WORKING PRINCIPLE:

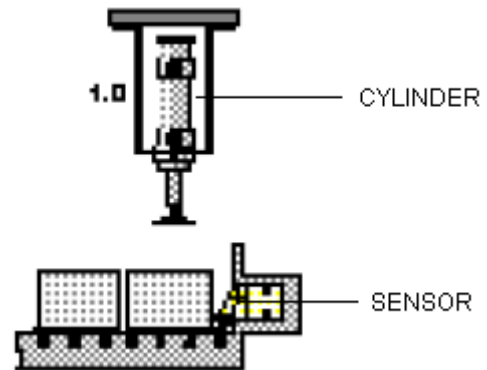
ELECTRO-PNEUMATIC CIRCUIT:

INFERENCE:

Exp. No.: 4B	Electro-pneumatic circuit for box lifting station.
Date:	

PROBLEM:

A transfer station removes a product from a conveyor belt. If the product is detected as present (By an electrical sensor) and if the operator presses an electrical push button, the pick-up cylinder advances. Upon release of the electrical push button, cylinder is to retract to the initial position. The transfer of the product should be happened in a controlled manner. Whereas the return stroke of the cylinder should be considerably fast. A 5/2 single solenoid spring return valve should be used.



WORKING PRINCIPLE:

ELECTRO-PNEUMATIC CIRCUIT:

INFERENCE:

EXERCISE – 5

Design, simulation and testing of electro-pneumatic circuits with multiple actuator with and without relay for various industrial applications.

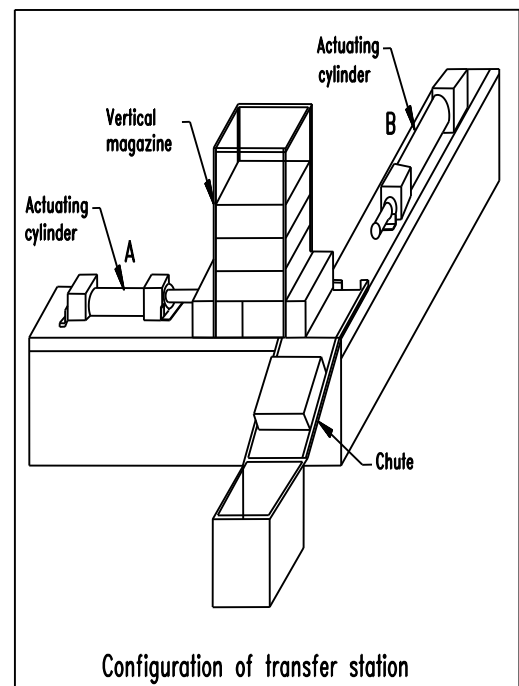
Exp. No.: 5 A	Electro-Pneumatic Circuit Design For Transfer Station
Date:	

PROBLEM:

Two cylinders are used to transfer parts from a magazine onto a chute. When a pushbutton is pressed, cylinder (A) extends, pushing the part from the magazine and positions it in preparation for transfer by cylinder (B) onto the out feed chute. Once the part is transferred, the cylinder (A) retracts, followed by the cylinder (B).

Sequence code: A1 B1 A0 B0

Logic Equation: $A1 = a_0 \cdot \text{Start}$; $B1 = a_1$; $A0 = b_1$; $B0 = a_0$

WORKING PRINCIPLE:

ELECTRO-PNEUMATIC CIRCUIT:

INFERENCE:

Exp. No.: 5B	Electro-Pneumatic Circuit Design For Transfer Station
Date:	

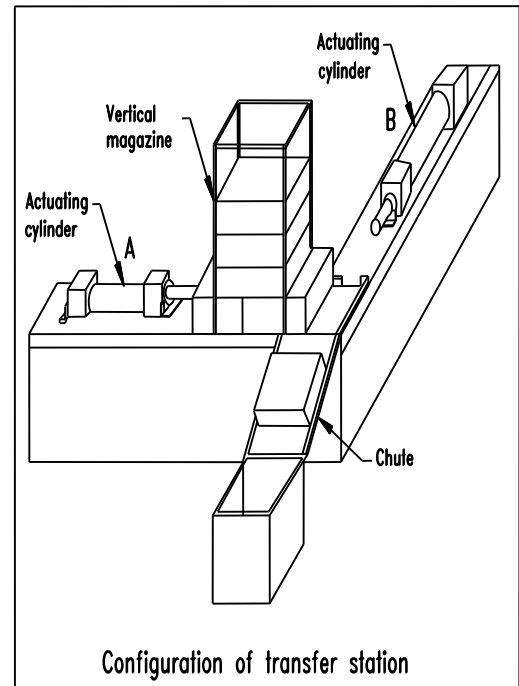
PROBLEM:

Two cylinders are used to transfer parts from a magazine onto a chute. When a pushbutton is pressed, cylinder (A) extends, pushing the part from the magazine and positions it in preparation for transfer by cylinder (B) onto the out feed chute. Once the part is transferred, the cylinder (B) retracts, followed by the cylinder (A).

Sequence code: A1 B1 B0 A0

Logic Equation: $A1 = a_0 \cdot \text{Start}$; $B1 = a_1$; $B0 = X_1$;
 $A0 = X_1 \cdot b_0$; $X_1 = b_1$; $X_0 = a_0$

WORKING PRINCIPLE:



ELECTRO-PNEUMATIC CIRCUIT:

INFERENCE:

EXERCISE – 6

Design and simulation of hydraulic circuits circuit using single acting and double acting cylinder

Exp. No.:6A

Date:

Actuation of single acting hydraulic cylinder

PROBLEM:

A single acting hydraulic cylinder is to be extended when a lever type 3/2 spring return hydraulic control valve is actuated. Until the lever is in operating condition, the single actuating cylinder should be in extended position and after the lever is back to original position the cylinder should retract to the home position. Use a unidirectional fixed displacement pump along with a pressure relief valve.

HYDRAULIC CIRCUIT:

WORKING PRINCIPLE:

INFERENCE:

Exp. No.:6 B	Actuation of Double Acting Hydraulic Cylinder
Date:	

PROBLEM:

Design a hydraulic double-acting cylinder circuit using a hydraulic valve with a configuration of 5/2 lever operated spring return valve. The cylinder is to extend when a lever is operated and when the lever is not operated due to spring the cylinder is to retract.

Design a hydraulic double-acting cylinder circuit using a hydraulic valve with a configuration of 5/3-way spring centered valve. The cylinder is to extend and return by actuating a separate levers for forward and return stroke.

WORKING PRINCIPLE:

HYDRAULIC CIRCUIT:

INFERENCE:

EXERCISE – 7

Design and simulation of hydraulic circuits using sequence valve and counterbalance valve applications.

Exp. No.:7A	Actuation of two cylinders with a help of pressure sequence valve
Date:	

PROBLEM:

Design a two-cylinder Hydraulic sequencing circuit using two pressure sequence valves which allow the automatic sequencing of two cylinders.

- The actuator-1 moves as soon as fluid flow is directed to the piston end of the actuator.
- Sequence valve blocks flow to the actuator-2 until a predetermined pressure is reached after the pressure sequence valve allows fluid flow to the actuator-2.
- The pre-determined pressure will be developed after the complete extension of cylinder-1.
- A sequence valve is typically fitted with an integral check valve. This allows freeflow of fluid around the valve when the direction of the actuator is reversed.
- After the full extension of cylinder-2 the oil goes through an another pressure sequence valve after the pre-determined pressure to move the cylinder-1 to the home position followed by the cylinder-2.

Use a hydraulic valve with a configuration of 4/3 way valve with spring centered. The valve is operated from both sides by a lever.

WORKING PRINCIPLE:

HYDRAULIC CIRCUIT:

INFERENCE:

Exp. No.:7 B	Control of hydraulic cylinder with the help of a counter balance valve
Date:	

PROBLEM:

A hydraulic cylinder is vertically mounted on which a load was mounted. Due to the applied load, cylinder may tend to extend automatically. If the load falls due to the extension of the cylinder automatically due to its self-weight, it leads to accident. Design a suitable hydraulic circuit to avoid the free falling of the hydraulic cylinder using a counter balance valve with check valve. A 4/3 way valve spring centered valve can be used. Both sides of the valve are operated by a lever.

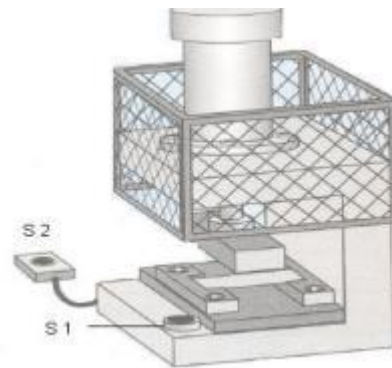
HYDRAULIC CIRCUIT:

WORKING PRINCIPLE:

INFERENCE:

EXERCISE – 8**Design and simulation of basic electro-hydraulic circuits****Exp. No.:8A****Date:****Control of hydraulic press using electro-hydraulic circuit along with a speed control****PROBLEM:**

The ram of a hydraulic press is only to extend if the protective guard is closed which actuates switch S1 and the start button switch S2 is operated. If either of these conditions is not met, the ram is to retract immediately. The double acting hydraulic cylinder is to be operated by a 4/2 single solenoid valve with spring return and speed of ram during the forward stroke should be controlled.

**HYDRAULIC CIRCUIT:**

WORKING PRINCIPLE:

INFERENCE:

Exp. No.:8B	Electro hydraulic circuit for speed control of hydraulic motor
Date:	

PROBLEM:

Design a hydraulic circuit to operate an industrial winch using a hydraulic motor. The hydraulic motor is used to move an industrial winch using a rope and rope winding mechanism. The upward and downward motion is due to the clockwise and anticlockwise movement of the motor along with the rope mechanism. The speed of the industrial winch should be controlled against the weight of the winch during the clockwise motion of the hydraulic motor (downward movement) by using counterbalance valve. Use a 4/3 way valve with spring centered hydraulic valve and operated by solenoids from both end. Employ a start switch to operate the winch and get confirmed from the electrical limit switch on both extreme end of the winch position.

HYDRAULIC CIRCUIT:

WORKING PRINCIPLE:**INFERENCE:**

EXERCISE – 9

Design and simulation of multiple actuator hydraulic and electro-hydraulic circuits for various industrial application with relay and without relay.

Exp. No.: 9A	Regenerative circuit of a double acting hydraulic cylinder
Date:	

PROBLEM:

Design a regenerative-cylinder-advance circuit, which provides a rapid advance to the point where a load is encountered. Fluid is recirculates from the rod end into the piston end and coupled with pump output to increase flow into the piston end, thus increasing cylinder extension speed. A 4/3 hydraulic control valve with the central position output and input is closed can be used. Another output is in blocked condition. A lever operates both sides of the valve. Drain port of the valve is directly connected to the tank.

HYDRAULIC CIRCUIT:

WORKING PRINCIPLE:**INFERENCE:**

Exp. No.: 9B	Design of multiple actuator hydraulic circuit using limit sensing valves.
Date:	

PROBLEM:

Two double acting hydraulic cylinder is to be operated with a sequence of A1 B1 A0 B0.

A1 – Represents the forward stroke of the hydraulic cylinder “A”

B1 – Represents the forward stroke of the hydraulic cylinder “B”

A0 – Represents the Return stroke of the hydraulic cylinder “A”

B0 – Represents the Return stroke of the hydraulic cylinder “B”

Use two 3/2 roller operated spring return hydraulic valve to sense the position of each hydraulic cylinders. A 3/2 lever operated spring return hydraulic valve should be used to start the sequence code. Ensure before starting the previous cycle was completed. Employ two 4/2 double pilot operated hydraulic valve to supply the oil to both cylinders.

WORKING PRINCIPLE:

HYDRAULIC CIRCUIT:

INFERENCE:

Ex. No: 10 A

Date:

Ladder programming for Boolean operations and math operations

Aim:

To construct a Ladder program for a given Boolean operation.

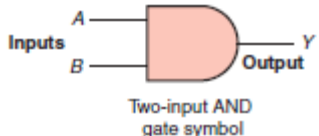
- OR, AND, NOR, NAND, EXOR

I/O Addressing:

S.No.	INPUT	ADDRESS	OUTPUT	ADDRESS
1.	A	%IX0.0	OR	%QX0.0
2.	B	%IX0.1	NOR	%QX0.1
3.			AND	%QX0.2
4.			NAND	%QX0.3
5.			EXOR	%QX0.4

Configuration:

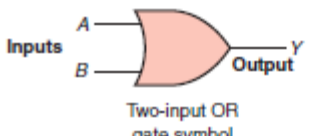
Logic Gates – Truth Table



Two-input AND gate symbol

AND truth table

Inputs		Output
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1



Two-input OR gate symbol

OR truth table

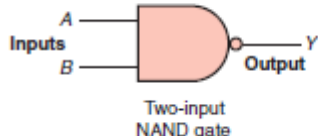
Inputs		Output
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1



NOT gate symbol

NOT truth table

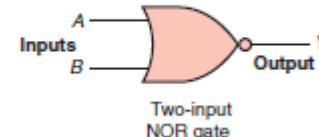
A	NOT A
0	1
1	0



Two-input NAND gate

NAND truth table

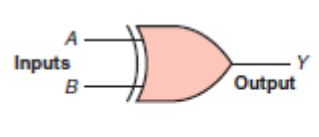
Inputs		Output
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0



Two-input NOR gate

NOR truth table

Inputs		Output
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0



Two-input EXOR gate

Truth table

Inputs		Output
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

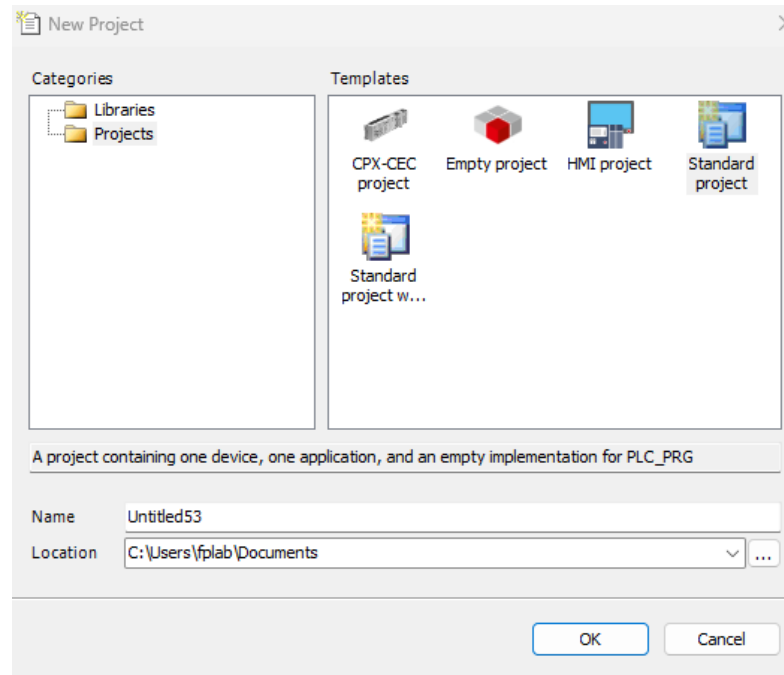
Sequence of operation:

STEP-1

Open Codesys software available on the machine

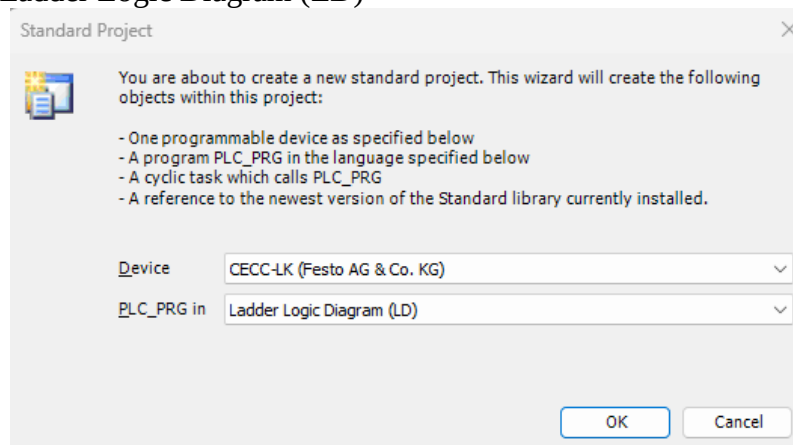
STEP-2

Click on the file New Project Standard project



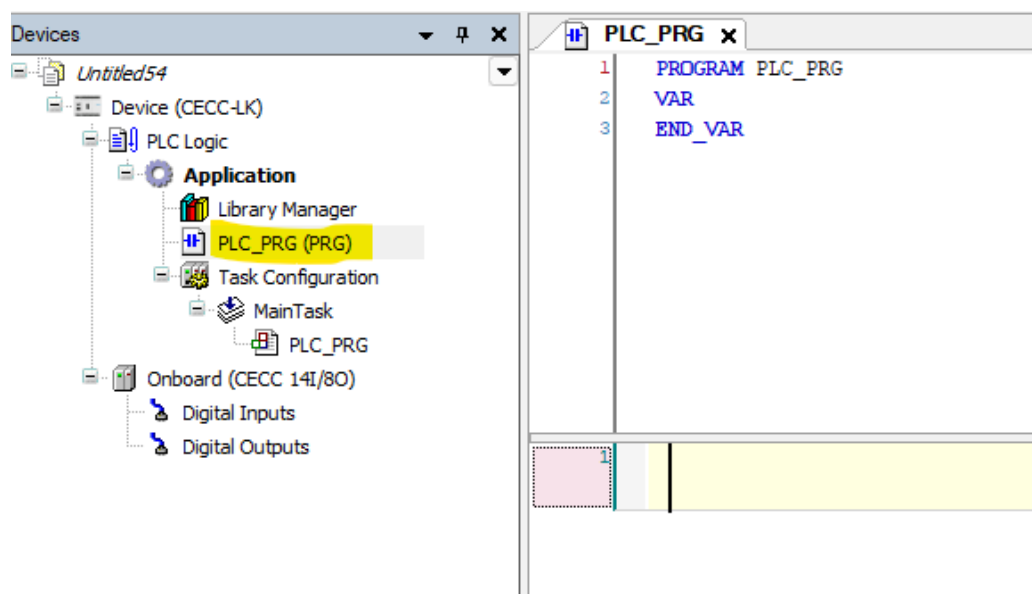
STEP-3

Select device as CECC-LK(Festo AG & Co,KG)
 PLC_PRG in → Ladder Logic Diagram (LD)



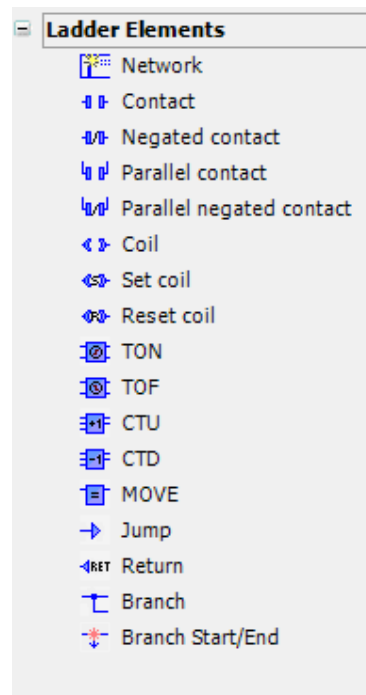
STEP-4

Make a double-click on PLC_PRG(PRG) which will open the ladder window

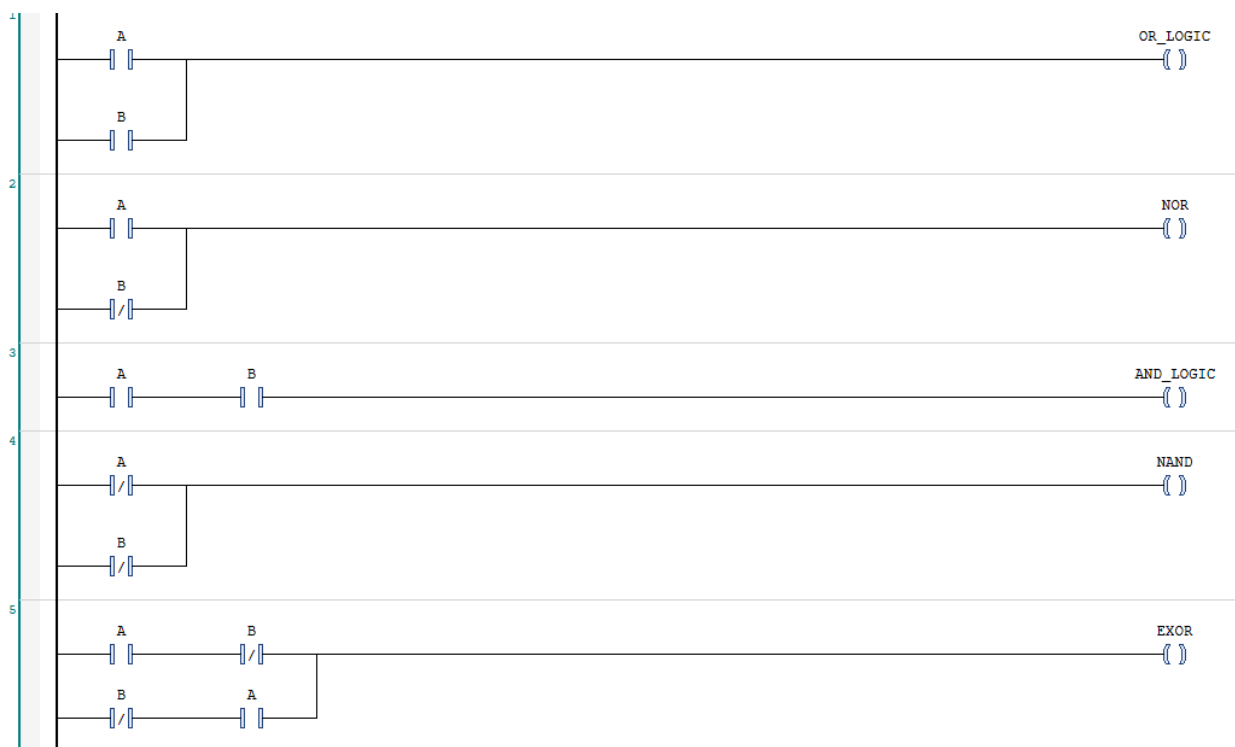


STEP-5

Use NO (Normally open), NC (Normally close), Parallel contact, coils from ladder elements from creating logical circuits

**STEP-6**

- Create the logical circuits for OR, AND, NOR, NAND, EXOR as shown below



- Provide variable address as shown below

Auto Declare

Scope: VAR

Name: A

Type: BOOL

Object: PLC_PRG [Application]

Initialization:

Address: %IX0.0|

Flags:

- ☐ CONSTANT
- ☐ RETAIN
- ☐ PERSISTENT

Comment:

OK Cancel

PROGRAM PLC_PRG

VAR

A AT %IX0.0: BOOL;

B AT %IX0.1: BOOL;

OR_LOGIC AT %QX0.0: BOOL;

NOR AT %QX0.1: BOOL;

AND_LOGIC AT %QX0.2: BOOL;

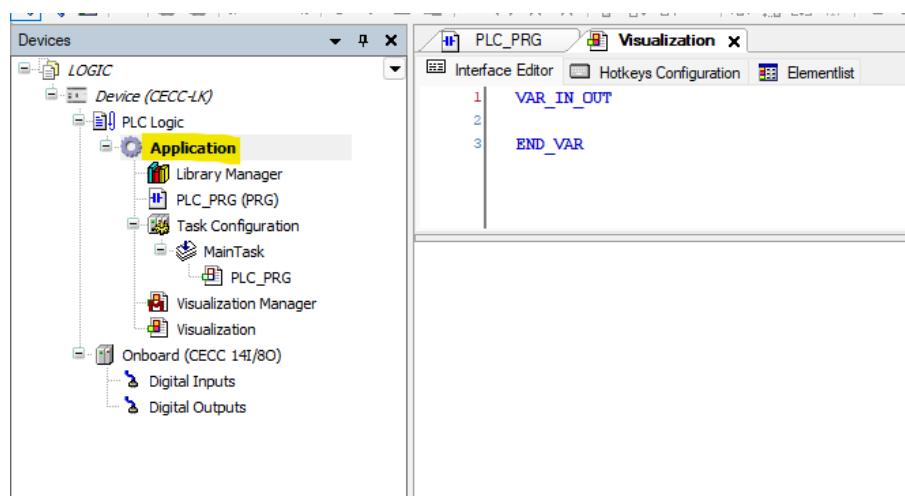
NAND AT %QX0.3: BOOL;

EXOR AT %QX0.4: BOOL;

END_VAR

STEP-7

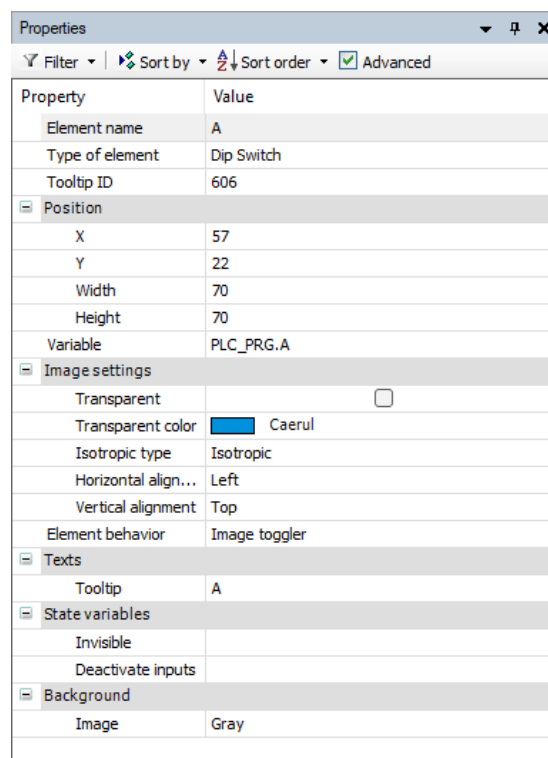
- Make a right-click on the application → Add object → Visualization



- Use the visualization toolbox to create a dipswitch with a lamp, as shown below.



- Change variables as mentioned in the Ladder to link both using properties



STEP-8

- Click on built → Rebuilt to verify the program
- To go into simulation mode use the online → Simulation
- Again use online → Login
- To debug the code use debug → Start
- Finally verify the Ladder using created switches and lamps in the visualization tab

Result:

Ex. No: 10 B**Date:****MATH OPERATIONS****Aim:**

To construct a Ladder program for a given math function (Addition, subtraction, multiplication, division, equal, not equal, less than or equal, greater than, greater than or equal)

I/O Addressing:

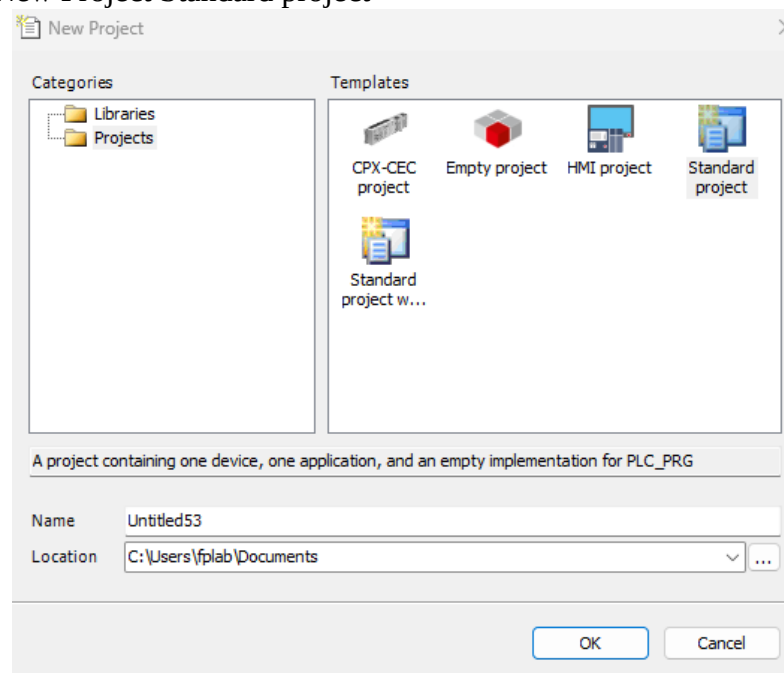
S.No.	INPUT	ADDRESS	OUTPUT	ADDRESS
1.	A	%IX0.0	Addition	%QX0.0
2.			Subtraction	%QX0.1
3.			Multiplication	%QX0.2
4.			Division	%QX0.3
5.			Equal	%QX0.4
6.			Not_Equal	%QX0.5
7.			Lessthan_OR_equal	%QX0.6
8.			Greater than	%QX0.7
9.			Greaterthan_or_equal	%QX0.8

Sequence of operation:**STEP-1**

Open codesys software available on the machine

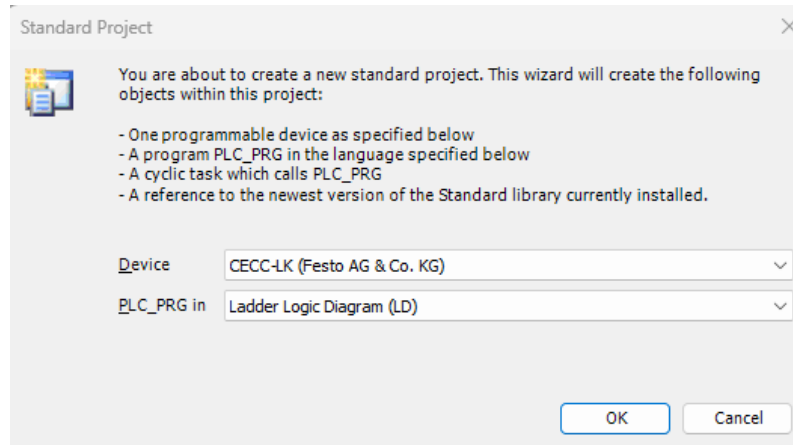
STEP-2

Click on the file New Project Standard project

**STEP-3**

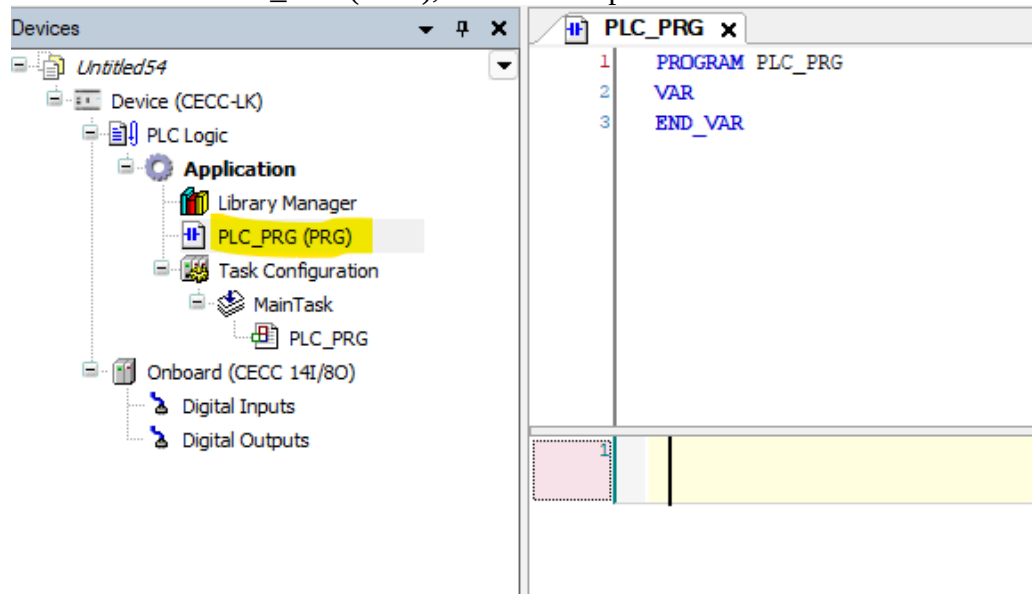
Select device as CECC-LK(Festo AG & Co,KG)

PLC_PRG in → Ladder Logic Diagram (LD)



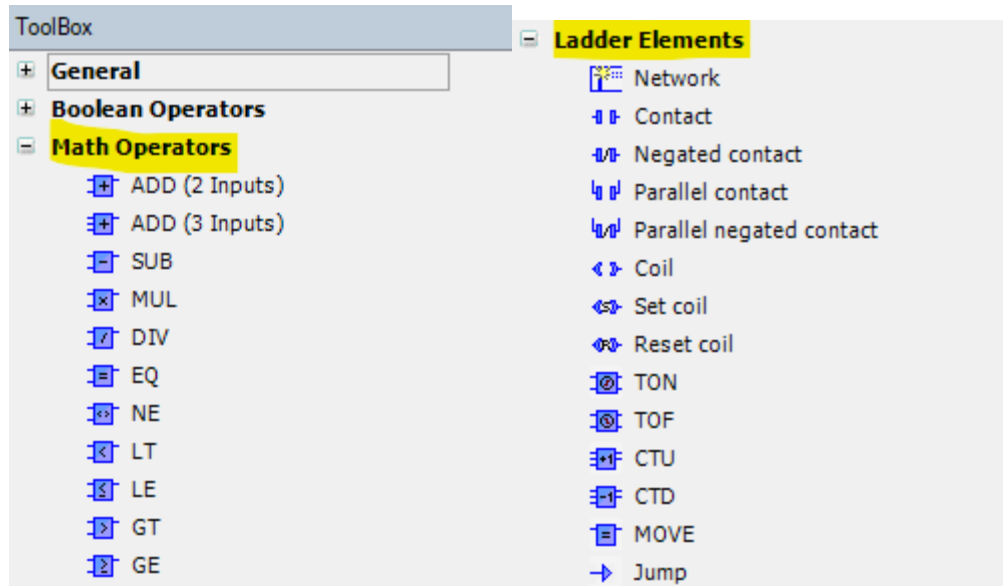
STEP-4

Make a double-click on PLC_PRG(PRG), which will open the ladder window.



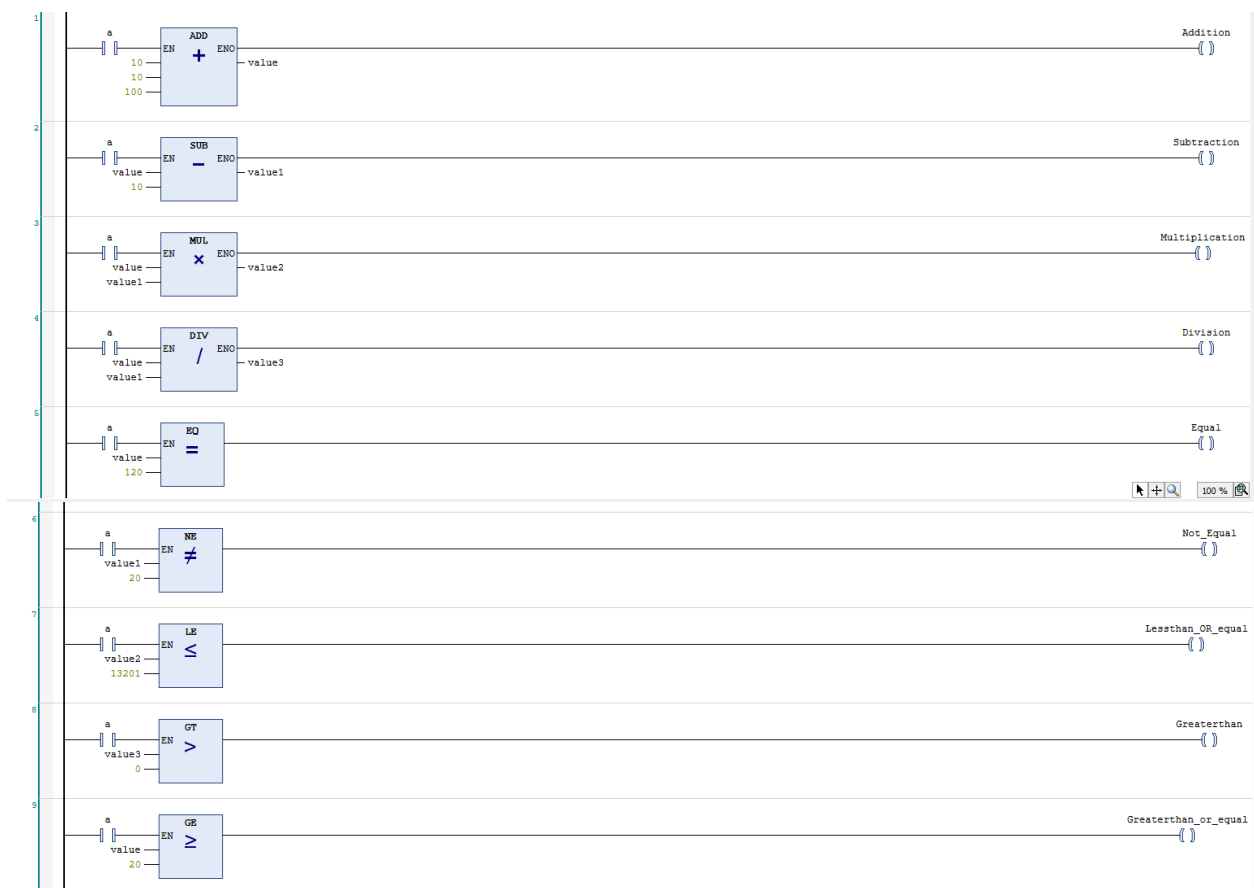
STEP-5

- Use NO (Normally open), coils from ladder elements, and Math operators for creating Addition, subtraction, multiplication, division, equal, not equal, less than or equal, greater than, greater than or equal.



STEP-6

- Create the math operation for Addition, subtraction, multiplication, division, equal, not equal, less than or equal, greater than, greater than or equal as shown below.



- Provide variable address as shown below

The 'Auto Declare' dialog box is shown with the following fields:

- Scope:** VAR
- Name:** A
- Type:** BOOL
- Object:** PLC_PRG [Application]
- Initialization:** (empty field with a browse button '...')
- Address:** %IX0.0
- Flags:**
 - ☐ CONSTANT
 - ☐ RETAIN
 - ☐ PERSISTENT
- Comment:** (empty text area)

Buttons at the bottom: OK, Cancel

PROGRAM PLC_PRG

VAR

```

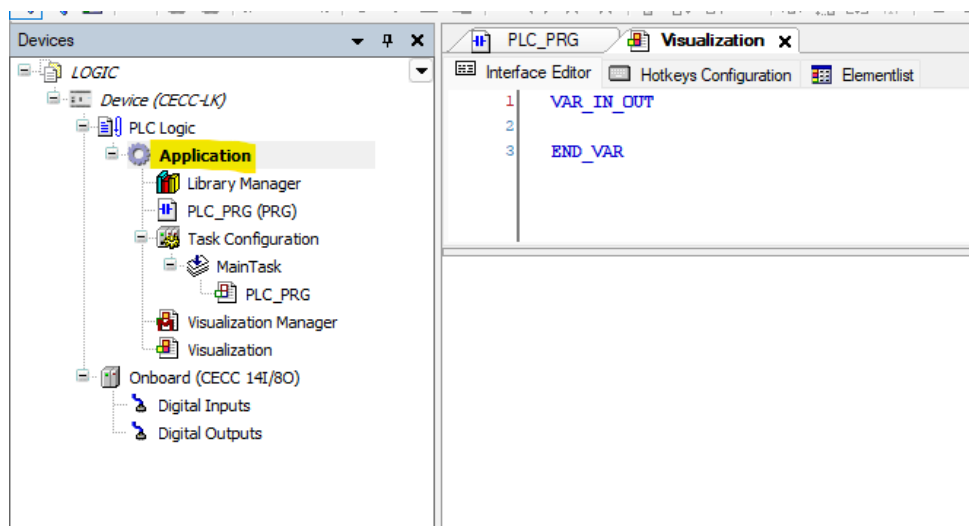
a: BOOL;
value: UINT;
value1: UINT;
value3: UINT;
value4: UINT;
value2: UINT;
Addition AT %QX0.0: BOOL;
Subtraction AT %QX0.1: BOOL;
Multiplication AT %QX0.2: BOOL;
Division AT %QX0.3: BOOL;
Equal AT %QX0.4: BOOL;
Not_Equal AT %QX0.5: BOOL;
Lessthan_OR_equal AT %QX0.6: BOOL;
Greaterthan AT %QX0.7: BOOL;
Greaterthan_or_equal AT %QX0.8: BOOL;

```

END_VAR

STEP-7

- Make a right-click on the application → Add object → Visualization



- Use visualization toolbox create dipswitch with lamp as shown below



- Change variables as mentioned in Ladder to link both using properties

Property	Value
Element name	A
Type of element	Dip Switch
Tooltip ID	606
Position	
X	57
Y	22
Width	70
Height	70
Variable	PLC_PRG.A
Image settings	
Transparent	☐
Transparent color	Caerul
Isotropic type	Isotropic
Horizontal align...	Left
Vertical alignment	Top
Element behavior	Image toggler
Texts	
Tooltip	A
State variables	
Invisible	
Deactivate inputs	
Background	
Image	Gray

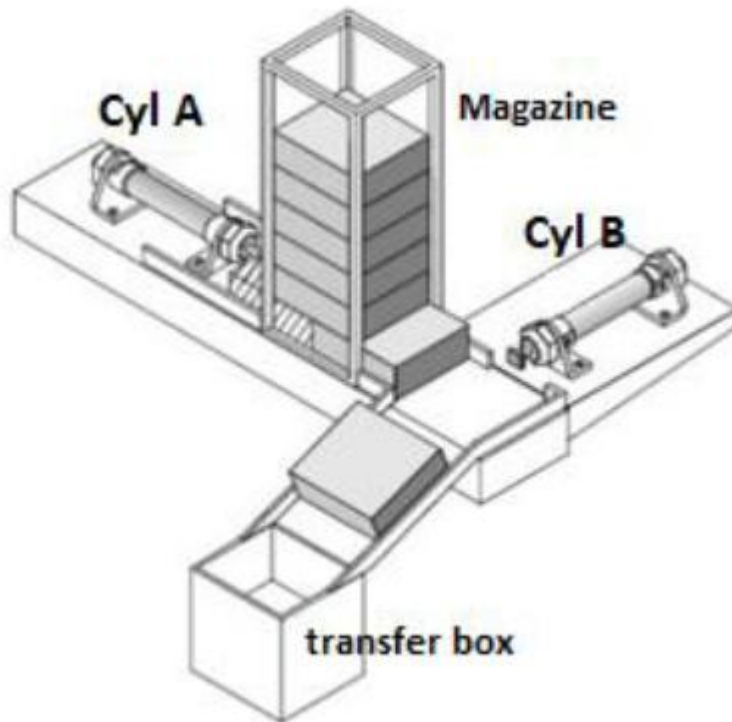
STEP-8

- Click on built → Rebuilt to verify the program
- To go into simulation mode use the online → Simulation
- Again use online → Login
- To debug the code use debug → Start
- Finally verify the Ladder using created switches and lamps in the visualization tab

Result:

Ex. No:11**Date:****INTERFACING OF ELECTRO – PNEUMATIC SYSTEM WITH PLC****Aim:**

A transfer system consists of a two-double-acting cylinder. Cylinder A will push part from the magazine, then back to its start position. The Cylinder B starts moving forward when the Cylinder A fully retracts. The Cylinder B only retracts when the component is fully transferred onto the transfer box. The cycle is to start when a start button is pressed. Design an Electro-pneumatic circuit for this application and test it with the kit.

**I/O Addressing:**

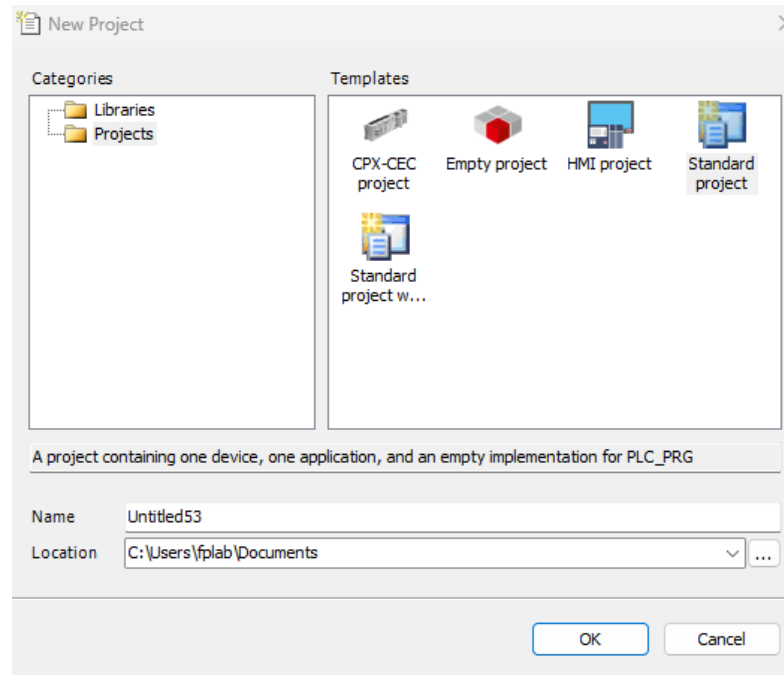
S.No.	INPUT	ADDRESS	OUTPUT	ADDRESS
1.	Start	%IX0.0	CylinderA_extend	%QX0.0
2.	Stop	%IX0.1	CylinderA_retract	%QX0.1
3.	A_retracted	%IX0.2	CylinderA_extend	%QX0.2
4.	A_extended	%IX0.3	CylinderA_retract	%QX0.3
5.	B_retracted	%IX0.4	Flag0	%MX0.0
6.	B_extended	%IX0.5	Flag1	%MX0.1

Sequence of operation:**STEP-1**

Open codesys software available on the machine.

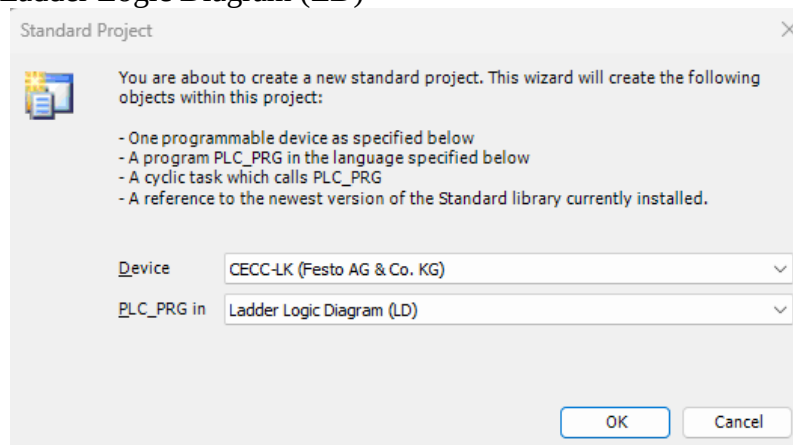
STEP-2

Click on the file → new project → Standard project.



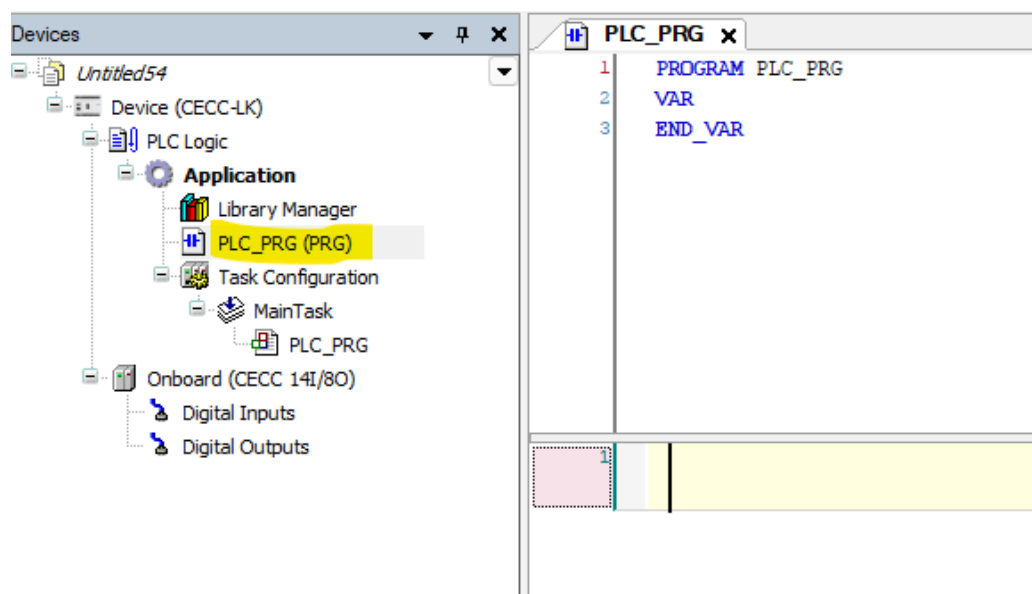
STEP-3

Select device as CECC-LK(Festo AG & Co,KG)
 PLC_PRG in → Ladder Logic Diagram (LD)



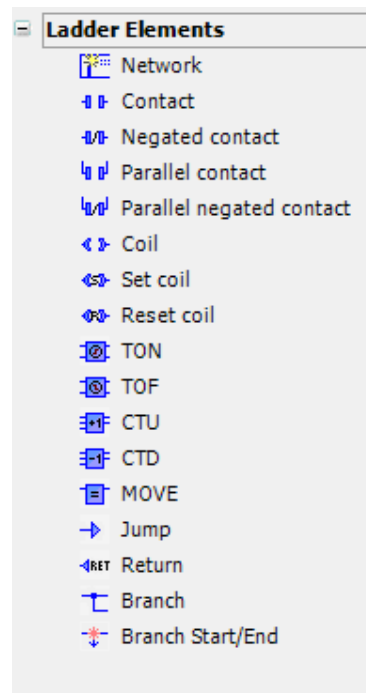
STEP-4

Double-click PLC_PRG(PRG), which will open the ladder window.

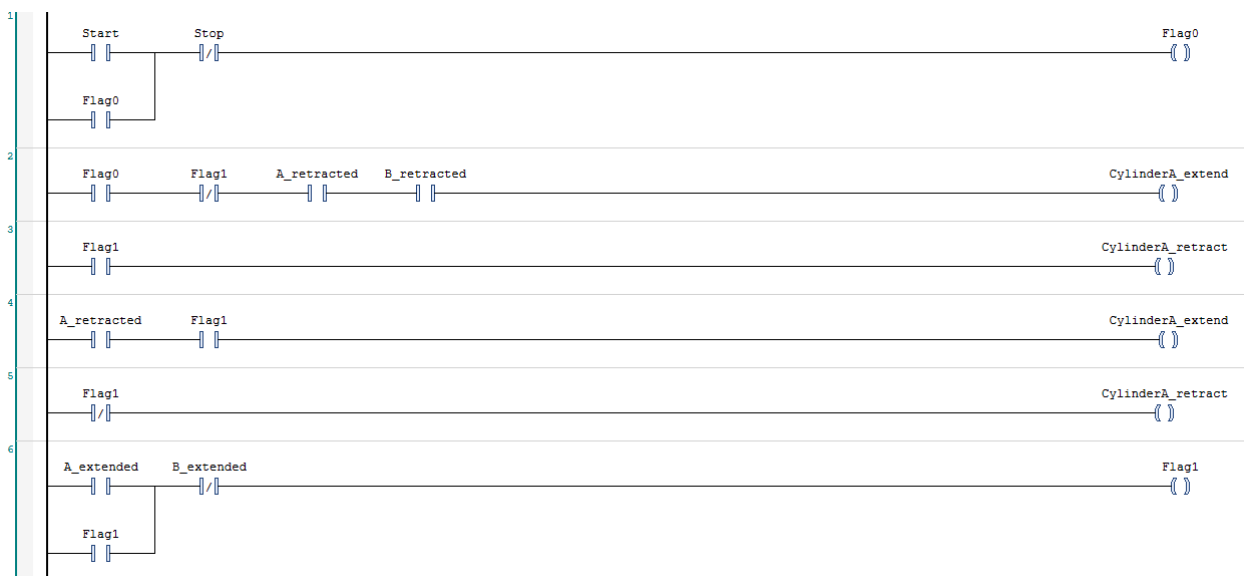


STEP-5

- Use NO (Normally open), NC (Normally close), and coils from ladder elements to create the sequence logic required.

**STEP-6**

- Complete the Ladder as shown below.



- Provide variable address as shown below

The 'Auto Declare' dialog box is shown with the following settings:

- Scope:** VAR
- Name:** A
- Type:** BOOL
- Object:** PLC_PRG [Application]
- Initialization:** (empty field with a browse button '...')
- Address:** %IX0.0
- Flags:**
 - ☐ CONSTANT
 - ☐ RETAIN
 - ☐ PERSISTENT
- Comment:** (empty text area)

Buttons: OK, Cancel

PROGRAM PLC_PRG

VAR

```

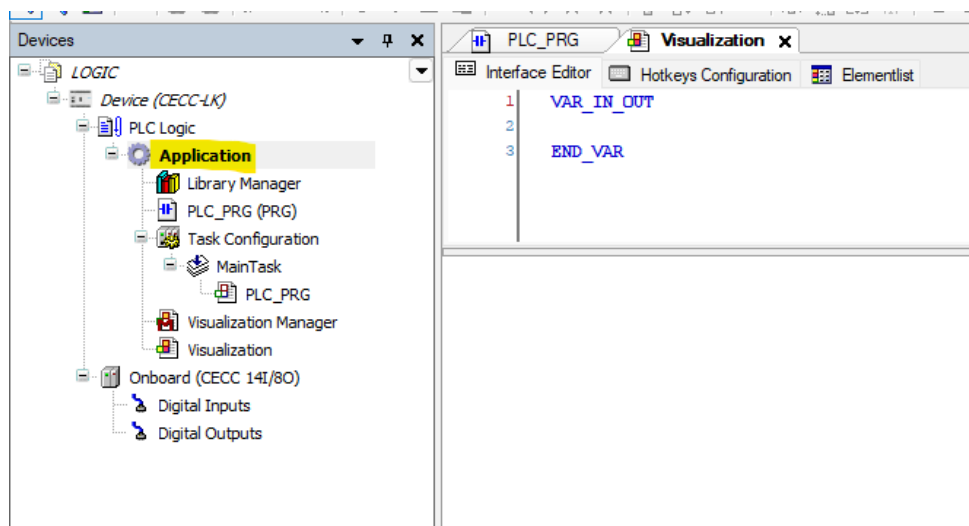
Start AT %IX0.0: BOOL;
Stop AT %IX0.1: BOOL;
Flag0 AT %MX0.0: BOOL;
A_retracted AT %IX0.2: BOOL;
A_extended AT %IX0.3: BOOL;
B_retracted AT %IX0.4: BOOL;
B_extended AT %IX0.5: BOOL;
CylinderA_extend AT %QX0.0: BOOL;
CylinderA_retract AT %QX0.1: BOOL;
CylinderB_extend AT %QX0.2: BOOL;
CylinderB_retract AT %QX0.3: BOOL;
Flag1 AT %MX0.1: BOOL;

```

END_VAR

STEP-7

- Make a right-click on the application → Add object → Visualization



- Use the visualization toolbox to create a dipswitch with a lamp, as shown below.



- Change variables as mentioned in Ladder to link both using properties.

Properties	
Filter Sort by Sort order Advanced	
Property	Value
Element name	A
Type of element	Dip Switch
Tooltip ID	606
Position	
X	57
Y	22
Width	70
Height	70
Variable	PLC_PRG.A
Image settings	
Transparent	☐
Transparent color	Caerul
Isotropic type	Isotropic
Horizontal align...	Left
Vertical alignment	Top
Element behavior	Image toggler
Texts	
Tooltip	A
State variables	
Invisible	
Deactivate inputs	
Background	
Image	Gray

STEP-8

- Click on built → Rebuilt to verify the program
- To go into simulation mode, use the online → Simulation
- Again, use online → Login
- To debug the code, use debug → Start
- Finally, verify the Ladder using created switches and lamps in the visualization tab.

Result:

Ex. No:12 A**Date:****PLC WITH TIMERS****Aim:**

Implement latching and unlatching of output with a particular time delay in PLC using a Ladder Diagram.

I/O Addressing:

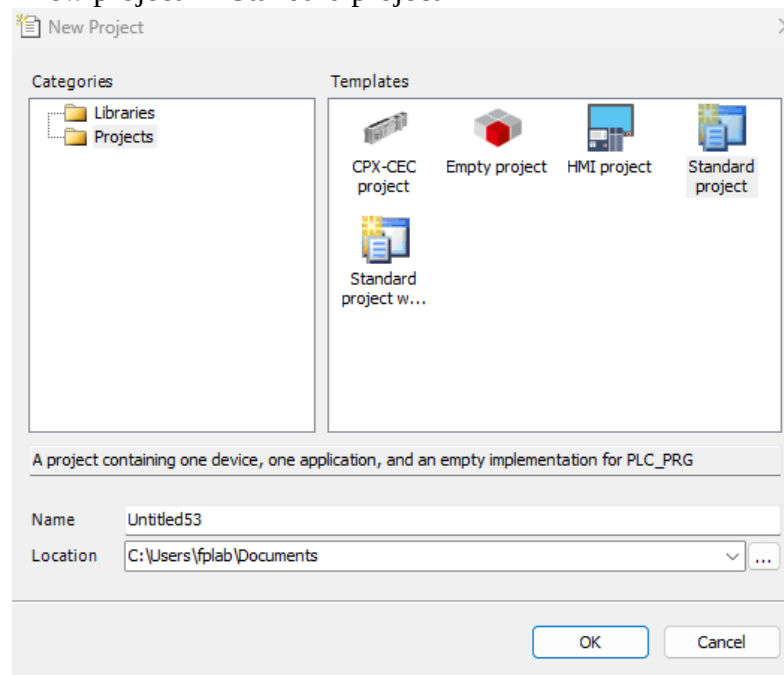
S.No.	INPUT	ADDRESS	OUTPUT	ADDRESS
1.	Start	%IX0.0	Cylinder_Coil	%QX0.0
2.	Stop	%IX0.1		
3.	Cylinder_retracted	%IX0.2		
4.	Cylinder_extended	%IX0.3		

Sequence of operation:**STEP-1**

Open codesys software available on the machine

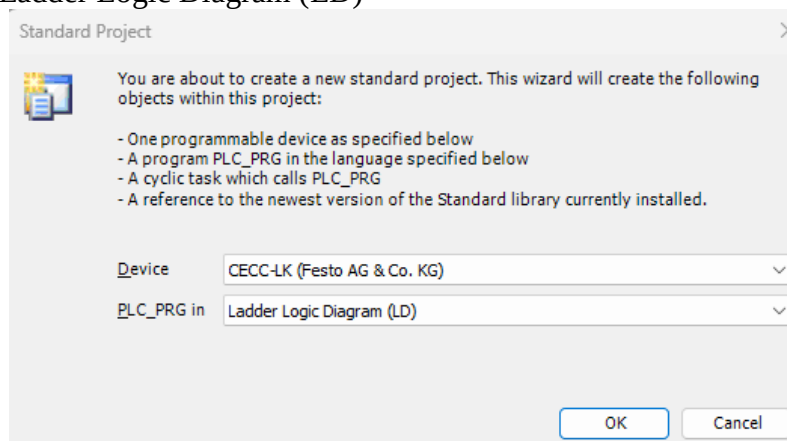
STEP-2

Click on the file → new project → Standard project

**STEP-3**

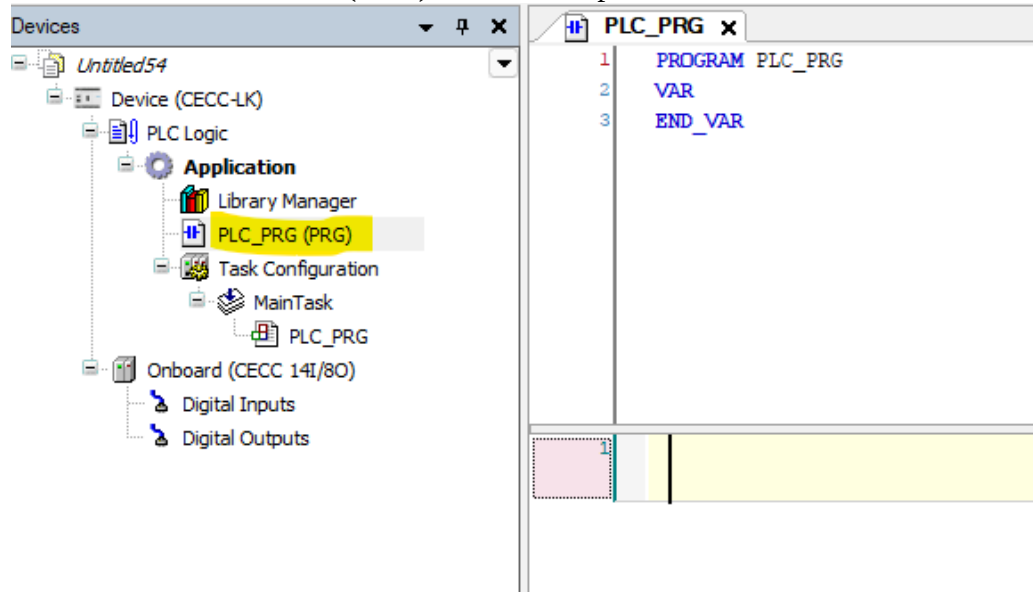
Select device as CECC-LK(Festo AG & Co,KG)

PLC_PRG in → Ladder Logic Diagram (LD)

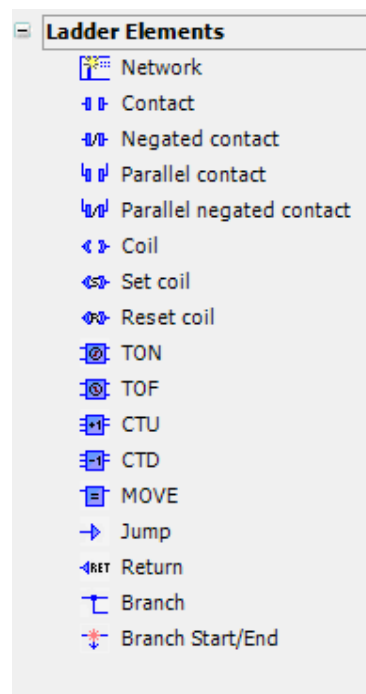


STEP-4

Make a double click on PLC_PRG(PRG) which will open ladder window

**STEP-5**

- Use NO (Normally open), NC (Normally close), coils and TON from ladder elements to create Timer circuit

**STEP-6**

- Complete the Ladder as shown below.



- Provide variable address as shown below

PROGRAM PLC_PRG

VAR

Cylinder_Coil AT %QX0.0: BOOL;

Start AT %IX0.0: BOOL;

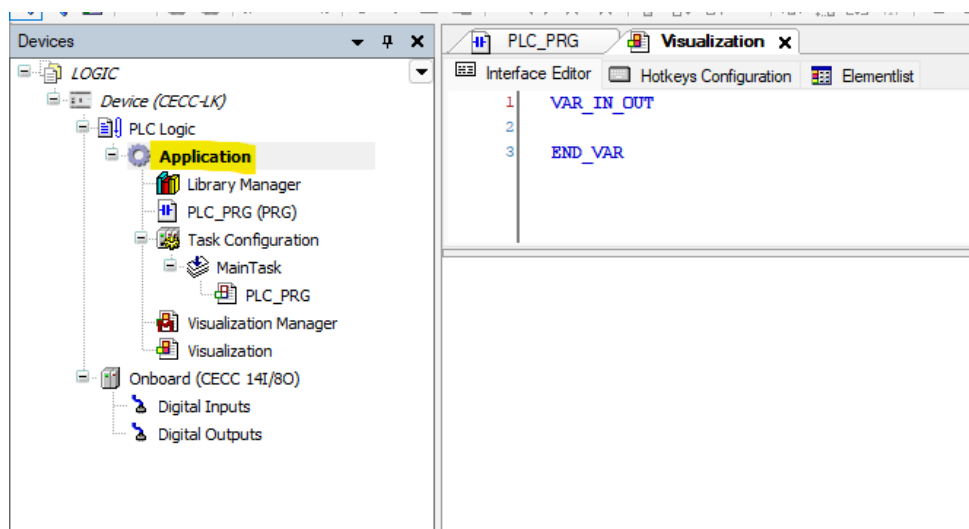
TON_1: TON;

Stop AT %IX0.1: BOOL;

END_VAR

STEP-7

- Make a right-click on the application → Add object → Visualization.



- Use the visualization toolbox to create a dipswitch with a lamp, as shown below.



- Change variables as mentioned in Ladder to link both using properties.

Properties	
Filter Sort by Sort order Advanced	
Property	Value
Element name	A
Type of element	Dip Switch
Tooltip ID	606
Position	
X	57
Y	22
Width	70
Height	70
Variable	PLC_PRG.A
Image settings	
Transparent	☐
Transparent color	Caerul
Isotropic type	Isotropic
Horizontal align...	Left
Vertical alignment	Top
Element behavior	Image toggler
Texts	
Tooltip	A
State variables	
Invisible	
Deactivate inputs	
Background	
Image	Gray

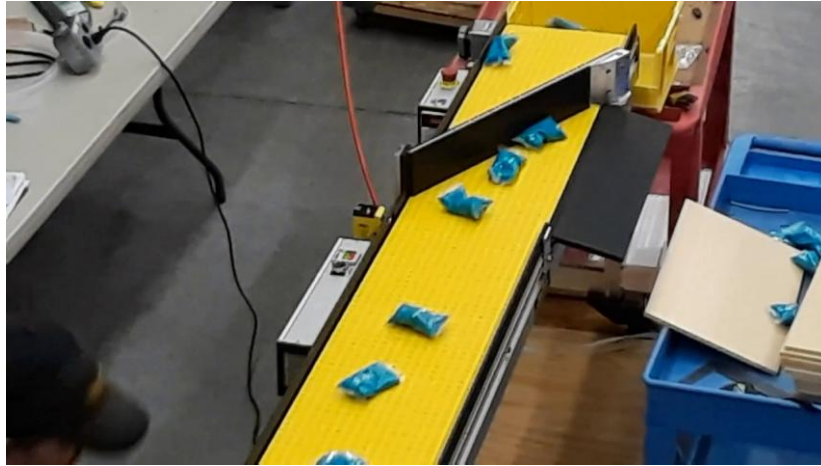
STEP-8

- Click on built → Rebuilt to verify the program
- To go into simulation mode, use the online → Simulation
- Again, use online → Login
- To debug the code, debug → Start
- Finally, verify the Ladder using the created switches and lamps in the visualization tab.

Result:

Ex. No:12 B**Date:****PLC WITH COUNTERS****Aim:**

Create a counter circuit for the conveyor unit available in the packaging area for each three-count splitter. The direction of the package should change. Use one single-acting package counting cylinder and one double-acting cylinder for the splitter circuit.

**I/O Addressing:**

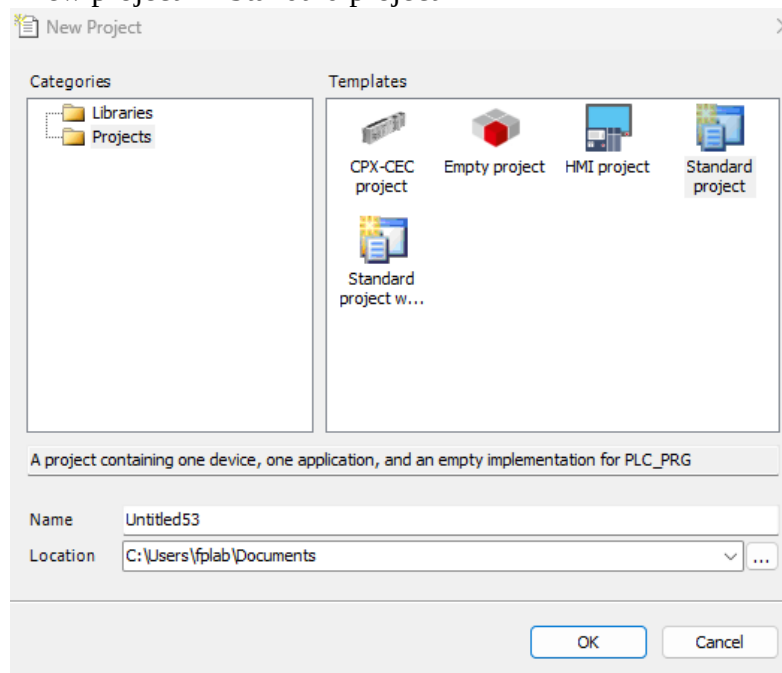
S.No.	INPUT	ADDRESS	OUTPUT	ADDRESS
1.	Start	%IX0.0	Counting_cylinder	%QX0.0
2.	Stop	%IX0.1	splitter_cylinder_extend	%QX0.1
3.	Cylinder_retracted	%IX0.2	splitter_cylinder_retract	%QX0.2
4.	Cylinder_extended	%IX0.3	Latch	%MX0.0

Sequence of operation:**STEP-1**

Open codesys software available in the machine

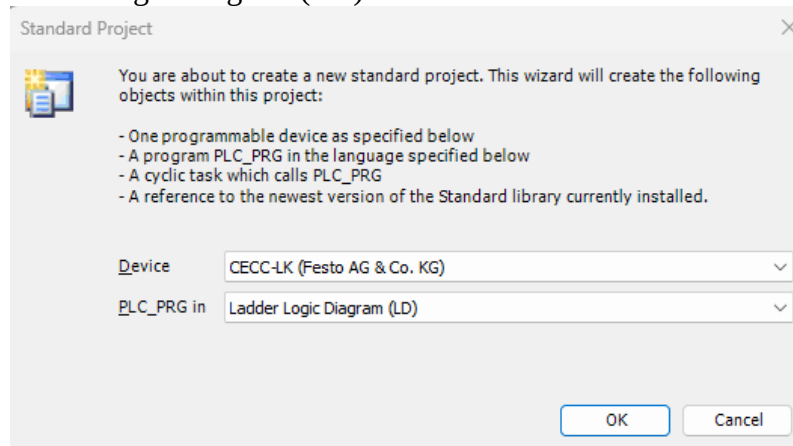
STEP-2

Click on the file → new project → Standard project

**STEP-3**

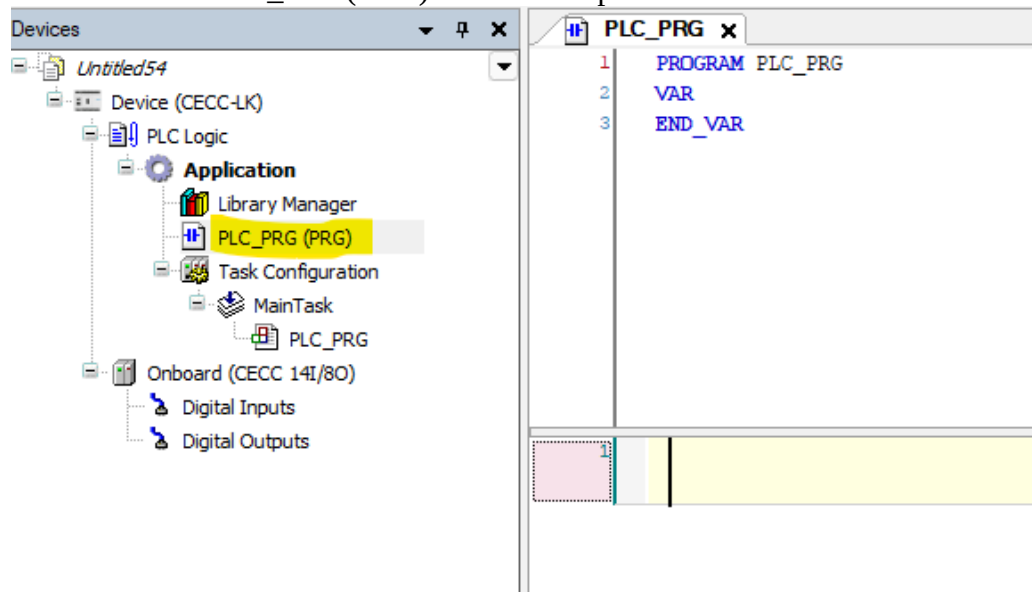
Select device as CECC-LK(Festo AG & Co,KG)

PLC_PRG in → Ladder Logic Diagram (LD)



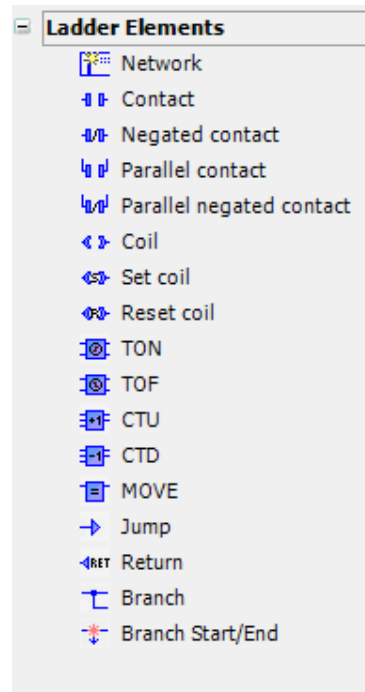
STEP-4

Make a double click on PLC_PRG(PRG) which will open ladder window



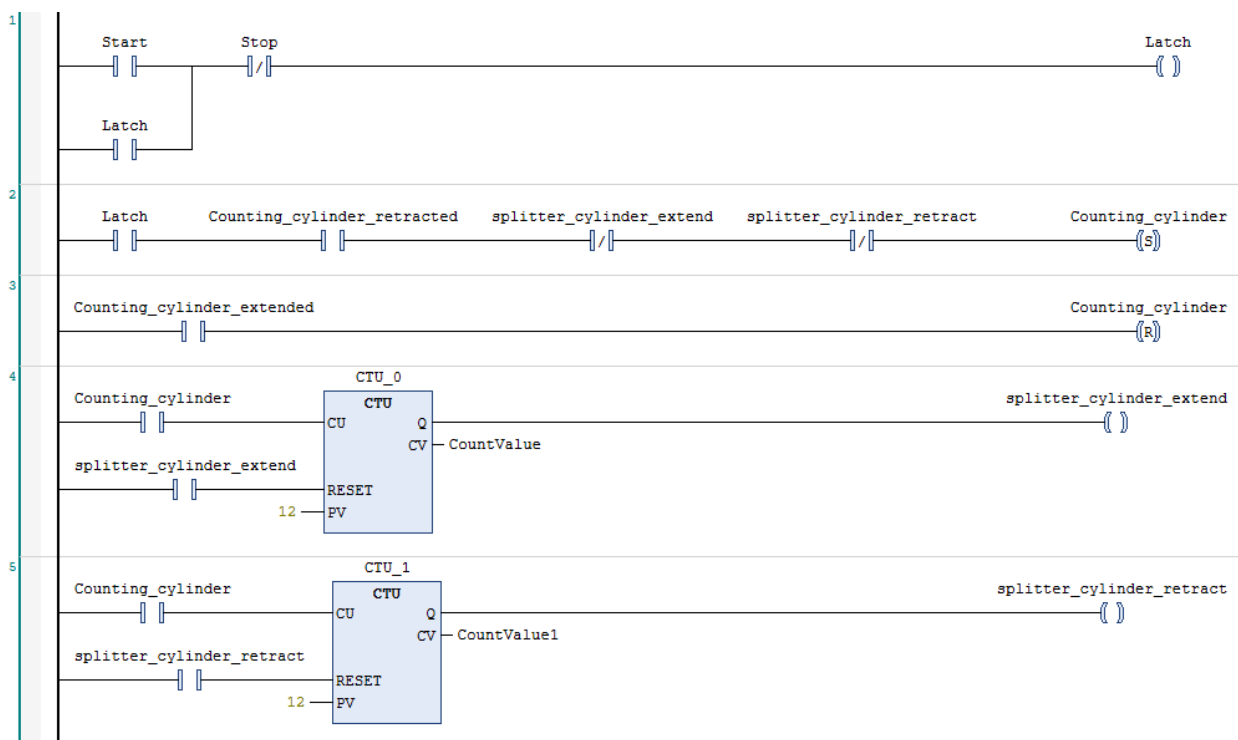
STEP-5

- Use NO (Normally open), NC (Normally close), coils and upcounter from ladder elements to create a given condition



STEP-6

- Complete the Ladder as shown below



- Provide variable address as shown below

The 'Auto Declare' dialog box is shown with the following settings:

- Scope:** VAR
- Name:** A
- Type:** BOOL
- Object:** PLC_PRG [Application]
- Initialization:** (empty)
- Address:** %IX0.0
- Flags:**
 - ☐ CONSTANT
 - ☐ RETAIN
 - ☐ PERSISTENT
- Comment:** (empty text area)

Buttons: OK, Cancel

PROGRAM PLC_PRG

VAR

```

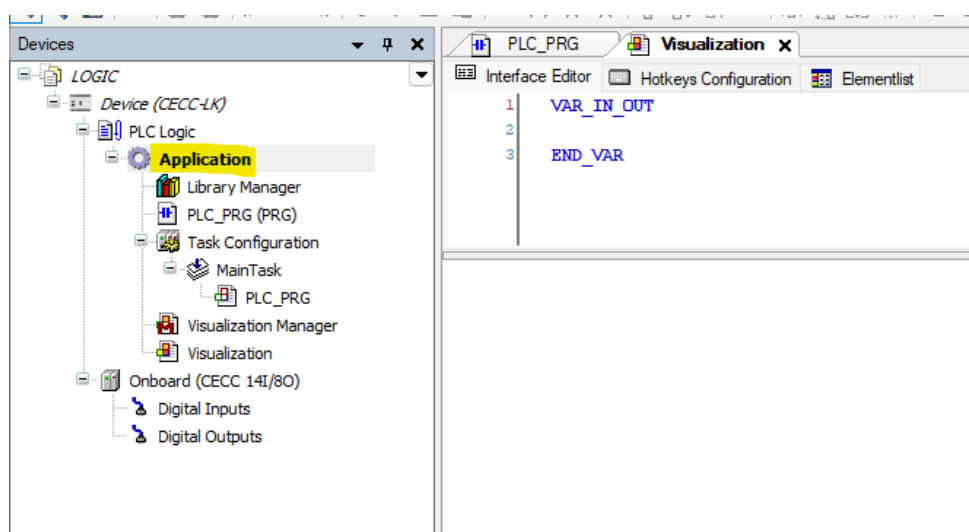
CTU: CTU;
CTU_0: CTU;
Start AT %IX0.0: BOOL;
Stop AT %IX0.1: BOOL;
Latch AT %MX0.0: BOOL;
Counting_cylinder AT %QX0.0: BOOL;
Counting_cylinder_retracted AT %IX0.2: BOOL;
Counting_cylinder_extended AT %IX0.3: BOOL;
CountValue: WORD;
CTU_1: CTU;
splitter_cylinder_extend AT %QX0.1: BOOL;
splitter_cylinder_retract AT %QX0.2: BOOL;

```

END_VAR

STEP-7

- Make a right-click on application→Add object→Visualization



- Use visualization toolbox create dipswitch with lamp as shown below



- Change variables as mentioned in Ladder to link both using properties

Properties	
Filter Sort by Sort order Advanced	
Property	Value
Element name	A
Type of element	Dip Switch
Tooltip ID	606
Position	
X	57
Y	22
Width	70
Height	70
Variable	PLC_PRG.A
Image settings	
Transparent	☐
Transparent color	Caerul
Isotropic type	Isotropic
Horizontal align...	Left
Vertical alignment	Top
Element behavior	Image toggler
Texts	
Tooltip	A
State variables	
Invisible	
Deactivate inputs	
Background	
Image	Gray

STEP-8

- Click on built → Rebuilt to verify the program
- To go into simulation mode use the online → Simulation
- Again use online → Login
- To debug the code use debug → Start
- Finally verify the Ladder using created switches and lamps in the visualization tab

Result:

Ex. No:13**Date:****MPS - DISTRIBUTION STATION****Aim:**

The Distributing station separates workpieces. Up to eight workpieces are stored in the Magazine tube of the stacking magazine. A double-acting cylinder pushes the workpieces out one at a time. The changer module grips the separated workpiece via a suction. The swivel arm of the changer, which is driven by a rotary actuator, moves the workpiece to the transfer point of the downstream station. Design a Mechatronics system for this application using an MPS distribution station.

Pin Allocation Table

Digital			
Function	SysLink	Color	Designation
I0	13	Grey-pink	Workpiece at beginning of conveyor
I1	14	Red-blue	Workpiece in middle of conveyor
I2	15	White-green	No workpiece at end of conveyor
I3	16	Brown-green	
I4	17	White-yellow	Slide retracted
I5	18	Brown-yellow	Slide advanced
I6	19	White-grey	Magazine empty
I7	20	Grey-brown	
Q0	1	White	Conveyor forward
Q1	2	Brown	Conveyor reverse
Q2	3	Green	Advance feed separator
Q3	4	Yellow	
Q4	5	Grey	Advance slide
Q5	6	Pink	
Q6	7	Blue	
Q7	8	Red	
24 V A	9+10	Black	24 V power supply for outputs
24 V B	21+22	White-pink	24 V power supply for inputs
GND A	11	Brown-pink	0 V power supply for outputs
GND A	12	Purple	0 V power supply for outputs
GND B	23+24	White-blue	0 V power supply for inputs

Control Console

Sensor	Input	Designation
PANEL_S1	E1.0	Start pushbutton
PANEL_S2	E1.1	Stop pushbutton (normally closed)
PANEL_S3	E1.2	Automatic/manual switch
PANEL_S4	E1.3	Reset pushbutton
Em_Stop	E1.5	EMERGENCY STOP unlatched
Actuator	Output	Designation
PANEL_P1	A1.0	Indicator light Start on
PANEL_P2	A1.1	Indicator light (Reset)
PANEL_P3	A1.2	Indicator light magazine empty
PANEL_P4	A1.3	

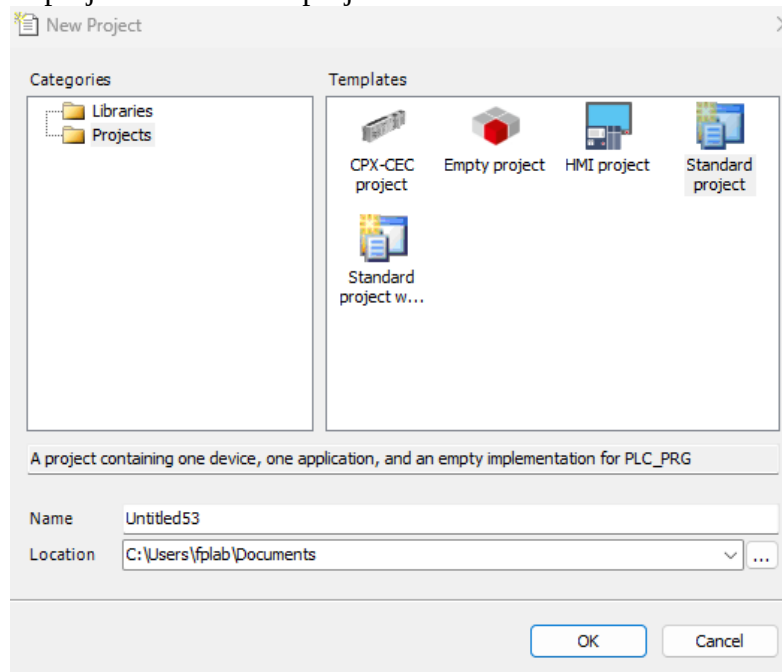
Sequence of operation:

STEP-1

Open codesys software available in the machine

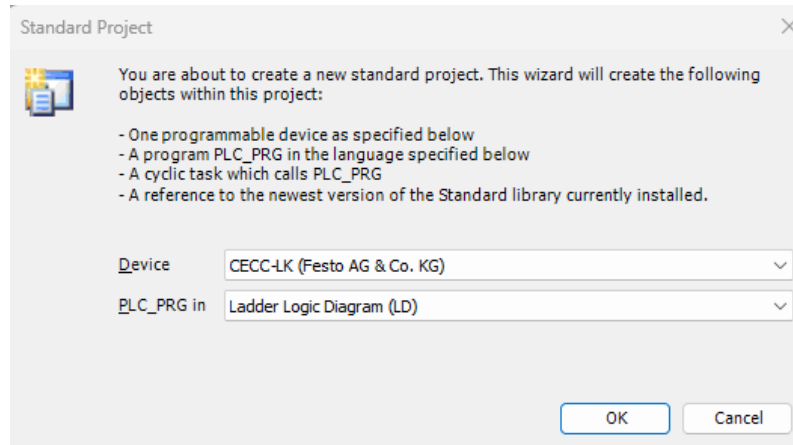
STEP-2

Click on file → new project → Standard project



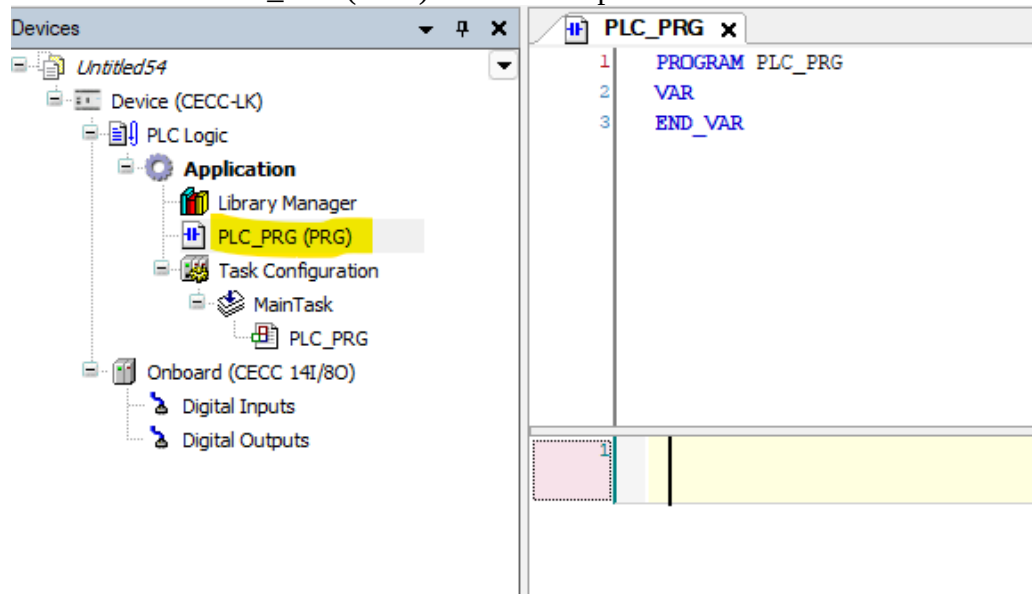
STEP-3

Select device as CECC-LK(Festo AG & Co,KG)
PLC_PRG in → Ladder Logic Diagram (LD)



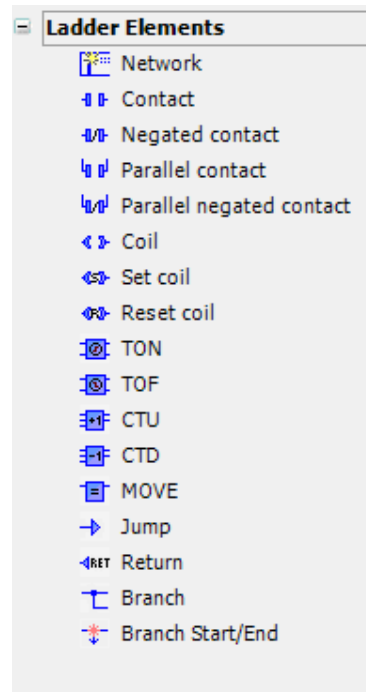
STEP-4

Make a double click on PLC_PRG(PRG) which will open ladder window

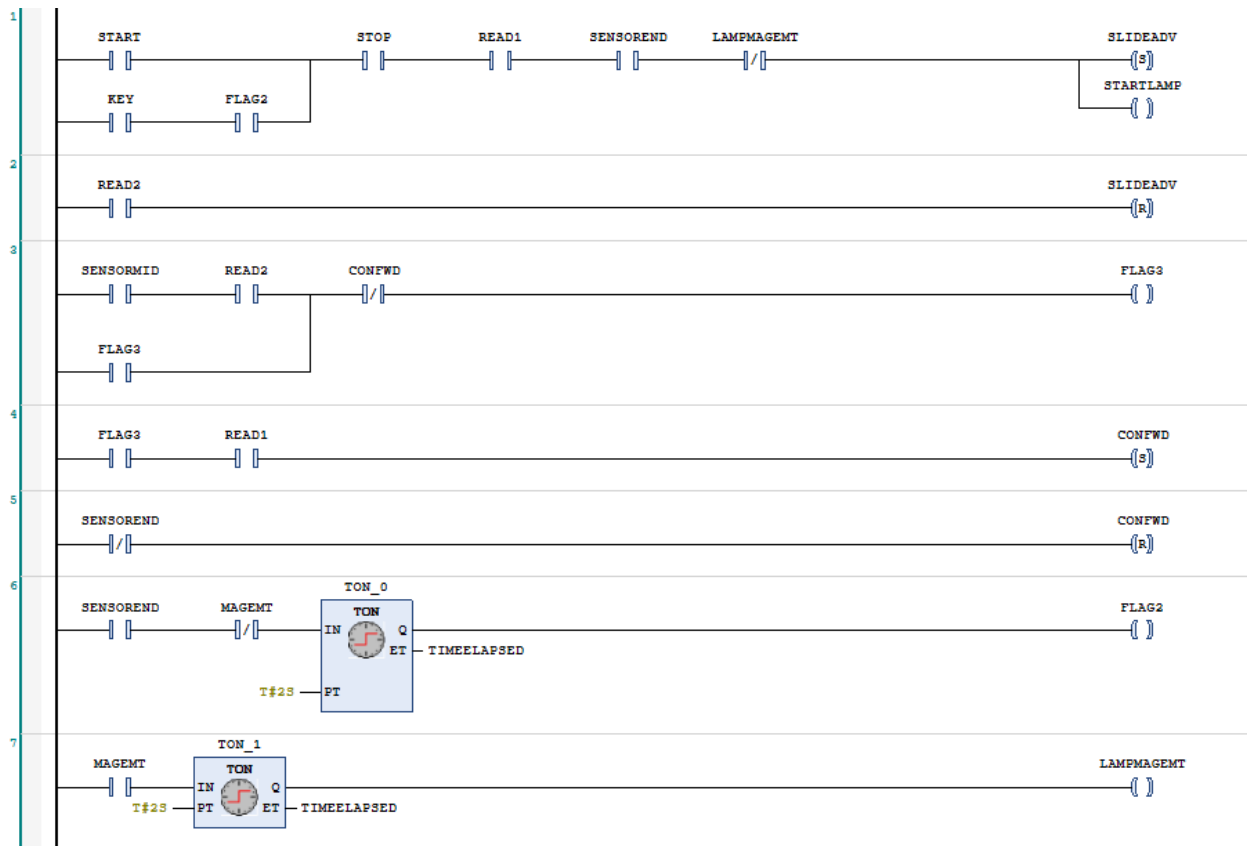


STEP-5

- Use NO (Normally open), NC (Normally close), coils, setcoil, resetcoil from ladder elements to create a given condition.

**STEP-6**

- Complete the Ladder as shown below.



- Provide variable address as shown below

Auto Declare

Scope	Name	Type
VAR	A	BOOL

Object: PLC_PRG [Application]

Initialization: ...

Address: %IX0.0

Flags:

- ☐ CONSTANT
- ☐ RETAIN
- ☐ PERSISTENT

Comment:

OK Cancel

PROGRAM PLC_PRG
VAR

```

START AT %IX2.0: BOOL;
STOP AT %IX2.1: BOOL;
STARTLAMP AT %QX1.0: BOOL;
SLIDEADV AT %QX0.4: BOOL;
READ1 AT %IX0.4: BOOL;
READ2 AT %IX0.5: BOOL;
SENSORMID AT %IX0.1: BOOL;
CONFWD AT %QX0.0: BOOL;
SENSOREND AT %IX0.2: BOOL;
MAGENT AT %IX0.6: BOOL;

```

```

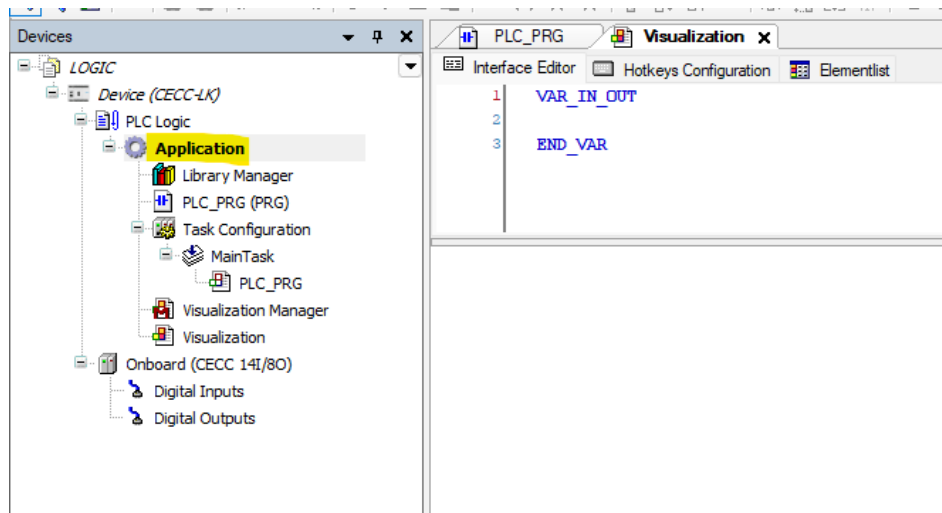
FLAG2 AT %MX0.1: BOOL;
TON_0: TON;
TIMEELAPSED: TIME;
KEY AT %IX2.2: BOOL;
FLAG3 AT %MX0.3: BOOL;
FLAG1 AT %MX0.0: BOOL;
LAMP MAGEMENT AT %QX1.2: BOOL;
TON_1: TON;

```

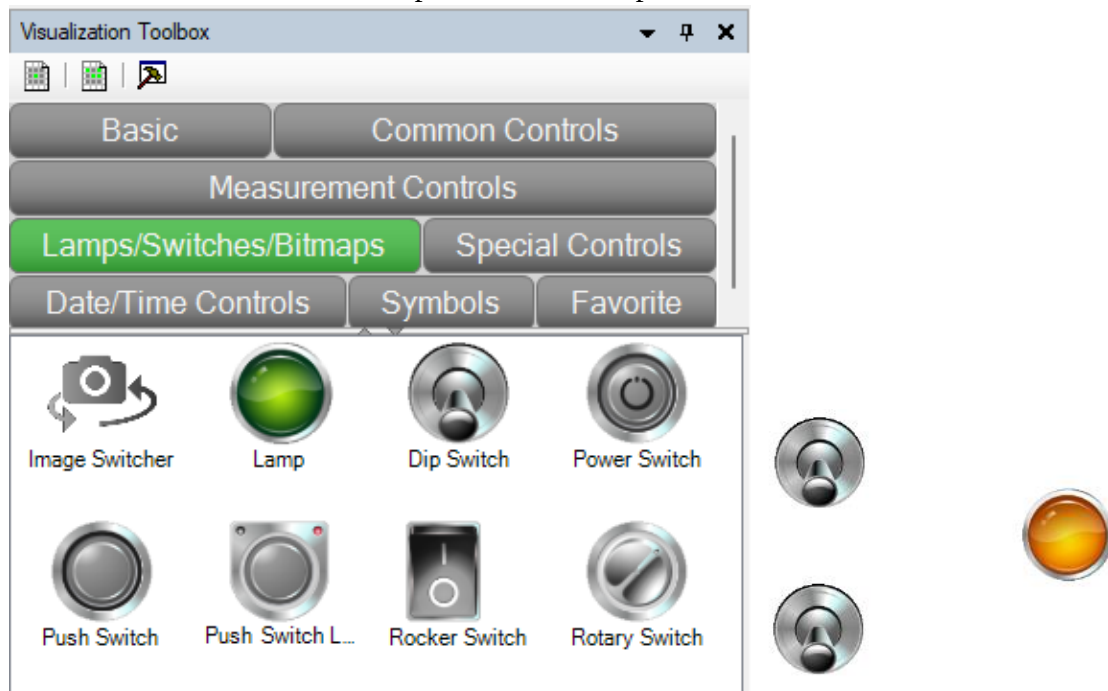
END_VAR

STEP-7

- Make a right click on application → Add object → Visualization



- Use visualization toolbox create dipswitch with lamp as shown below



- Change variables as mentioned in Ladder to link both using properties

Properties	
Filter Sort by Sort order Advanced	
Property	Value
Element name	A
Type of element	Dip Switch
Tooltip ID	606
Position	
X	57
Y	22
Width	70
Height	70
Variable	PLC_PRG.A
Image settings	
Transparent	☐
Transparent color	Caerul
Isotropic type	Isotropic
Horizontal align...	Left
Vertical alignment	Top
Element behavior	Image toggler
Texts	
Tooltip	A
State variables	
Invisible	
Deactivate inputs	
Background	
Image	Gray

STEP-8

- Click on built → Rebuilt to verify the program
- To go into simulation mode use the online → Simulation
- Again use online → Login
- To debug the code use debug → Start
- Finally verify the Ladder using created switches and lamps in the visualization tab

Result:

Ex. No:14**Date:****MPS - SORTING STATION****Aim:**

Workpieces are sorted into three slides by the sorting station. A diffusion sensor detects the workpieces positioned at the conveyor's beginning. The workpiece's properties (black, red, metal) are detected by sensors upstream of the stopper. Workpieces can be sorted into the proper slides using sorting gates that are operated by short-stroke cylinders through the use of a deflector. A sensor that is retroreflective keeps track of the slides' level. Using an MPS sorting station, design a mechatronic system.

Pin Allocation Table

Function	SysLink	Color	Designation
I0	13	Grey-pink	Workpiece at beginning of conveyor
I1	14	Red-blue	Deflector 1 extended
I2	15	White-green	Slide full
I3	16	Brown-green	Deflector 2 extended
I4	17	White-yellow	Workpiece detected
I5	18	Brown-yellow	Workpiece not black
I6	19	White-grey	Workpiece metallic
I7	20	Grey-brown	
Q0	1	White	Conveyor forward
Q1	2	Brown	Extend deflector 1
Q2	3	Green	Extend deflector 2
Q3	4	Yellow	Retract stopper
Q4	5	Grey	
Q5	6	Pink	
Q6	7	Blue	
Q7	8	Red	
24 V A	9+10	Black	24 V power supply for outputs
24 V B	21+22	White-pink	24 V power supply for inputs
GND A	11	Brown-pink	0 V power supply for outputs
GND A	12	Purple	0 V power supply for outputs
GND B	23+24	White-blue	0 V power supply for inputs

Control Console

Sensor	Input	Designation
PANEL_S1	E1.0	Start pushbutton
PANEL_S2	E1.1	Stop pushbutton (normally closed)
PANEL_S3	E1.2	Automatic/manual switch
PANEL_S4	E1.3	Reset pushbutton
Em_Stop	E1.5	EMERGENCY STOP unlatched
Actuator	Output	Designation
PANEL_P1	A1.0	Indicator light Start on
PANEL_P2	A1.1	Indicator light (Reset)
PANEL_P3	A1.2	Indicator chute full
PANEL_P4	A1.3	

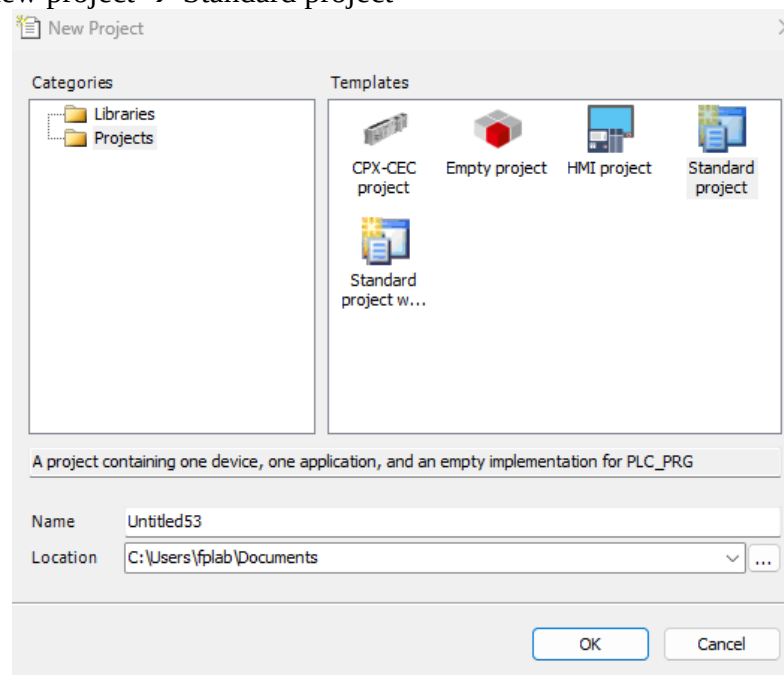
Sequence of operation:

STEP-1

Open codesys software available in the machine

STEP-2

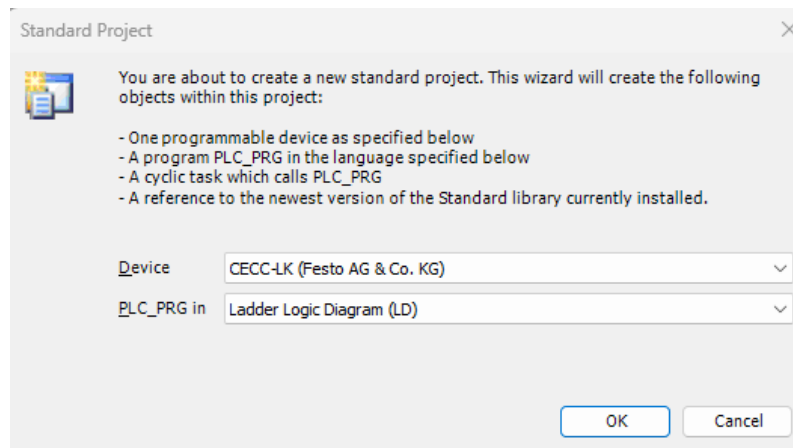
Click on file → new project → Standard project



STEP-3

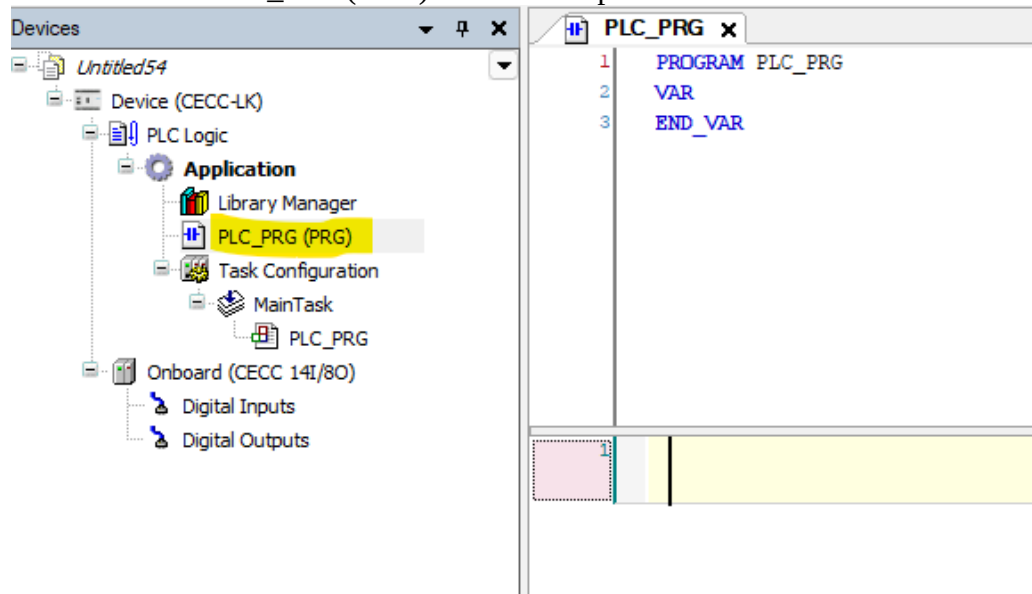
Select device as CECC-LK(Festo AG & Co,KG)

PLC_PRG in → Ladder Logic Diagram (LD)



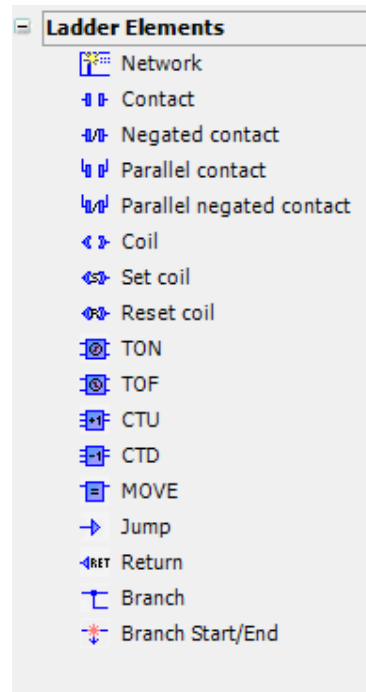
STEP-4

Make a double click on PLC_PRG(PRG) which will open ladder window



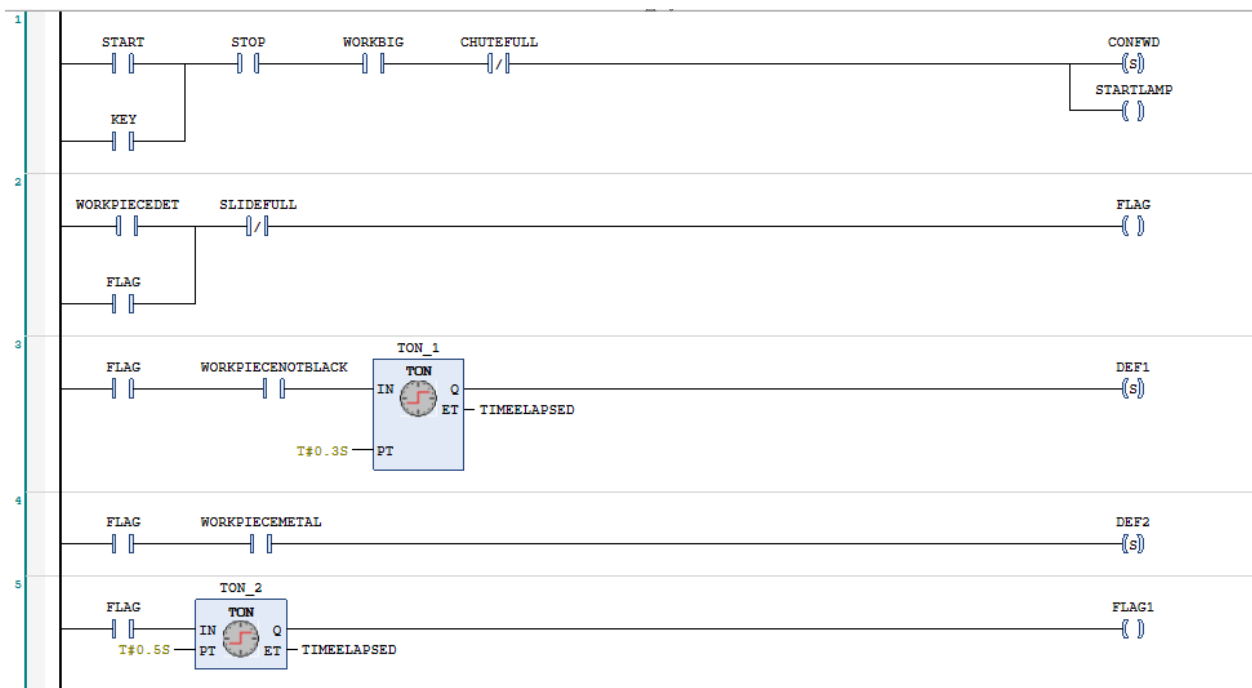
STEP-5

- Use NO (Normally open), NC (Normally close), coils, setcoil, resetcoil from ladder elements to create a given condition



STEP-6

- Complete the Ladder as shown below





- Provide variable address as shown below

Auto Declare

Scope	Name	Type
VAR	A	BOOL

Object	Initialization	Address
PLC_PRG [Application]		%IX0.0

Flags

☐ CONSTANT

☐ RETAIN

☐ PERSISTENT

Comment

OK Cancel

PROGRAM PLC_PRG

VAR

```

START AT %ix1.0: BOOL;
STARTLAMP AT %QX1.0: BOOL;
STOP AT %IX1.1: BOOL;
CONFWD AT %QX0.0: BOOL;
WORKBIG AT %IX0.0: BOOL;
RESET AT %IX1.3: BOOL;
DEF1 AT %QX0.1: BOOL;
DEF1SIGNAL AT %IX0.1: BOOL;
STOPPER AT %QX0.3: BOOL;
WORKPIECEDET AT %IX0.4: BOOL;
WORKPIECENOTBLACK AT %IX0.5: BOOL;
SLIDEFULL AT %IX0.2: BOOL;

```

```

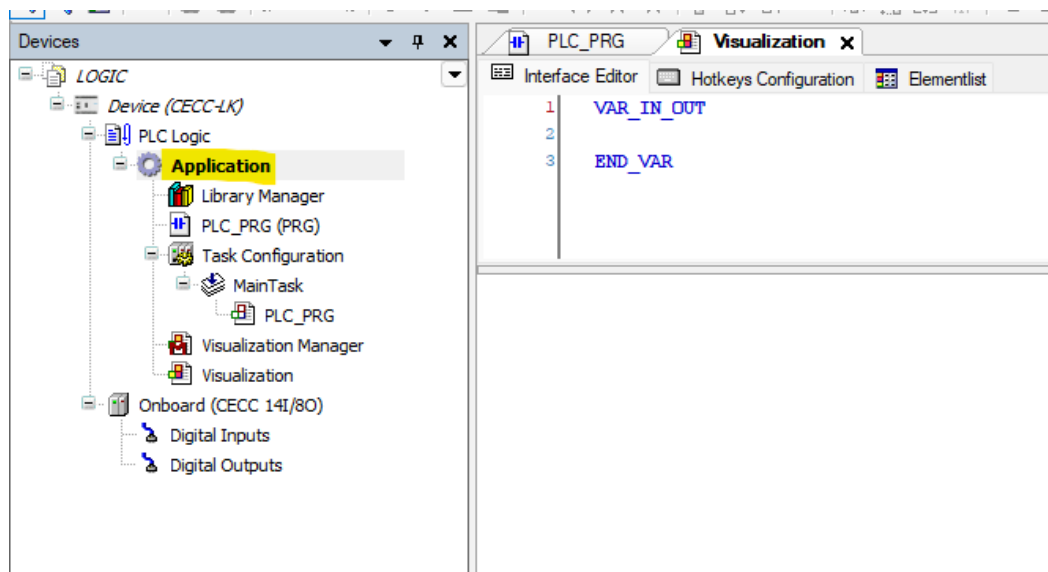
DEF2SIGNAL AT %IX0.3: BOOL;
WORKPIECMETAL AT %IX0.6: BOOL;
DEF2 AT %QX0.2: BOOL;
TON_0: TON;
TIMEELAPSED: TIME;
FLAG1 AT %MX0.0: BOOL;
TON_1: TON;
FLAG AT %MX0.5: BOOL;
TON_2: TON;
KEY AT %IX1.2: BOOL;
TON_3: TON;
CHUTEFULL AT %QX1.2: BOOL;
RESETLAMP AT %QX1.1: BOOL;

```

END_VAR

STEP-7

- Make a right click on application→Add object→Visualization



- Use visualization toolbox create dipswitch with lamp as shown below



- Change variables as mentioned in Ladder to link both using properties

Properties	
Filter Sort by Sort order Advanced	
Property	Value
Element name	A
Type of element	Dip Switch
Tooltip ID	606
Position	
X	57
Y	22
Width	70
Height	70
Variable	PLC_PRG.A
Image settings	
Transparent	☐
Transparent color	Caerul
Isotropic type	Isotropic
Horizontal align...	Left
Vertical alignment	Top
Element behavior	Image toggler
Texts	
Tooltip	A
State variables	
Invisible	
Deactivate inputs	
Background	
Image	Gray

STEP-8

- Click on built → Rebuilt to verify the program
- Go into simulation mode use the online → Simulation
- Again use online → Login
- To debug the code use debug → Start
- Finally verify the Ladder using created switches and lamps in the visualization tab

Result:

Ex. No: 15**Date:**

DESIGNING THE SETUP FOR A FACTORY AND PROGRAMMING THE PLC WITH VARIOUS APPLICATIONS

Aim:

To design the setup for a factory and program the PLC with various applications.

Equipment Required:

PC with codesys software and factory IO software

Theory:

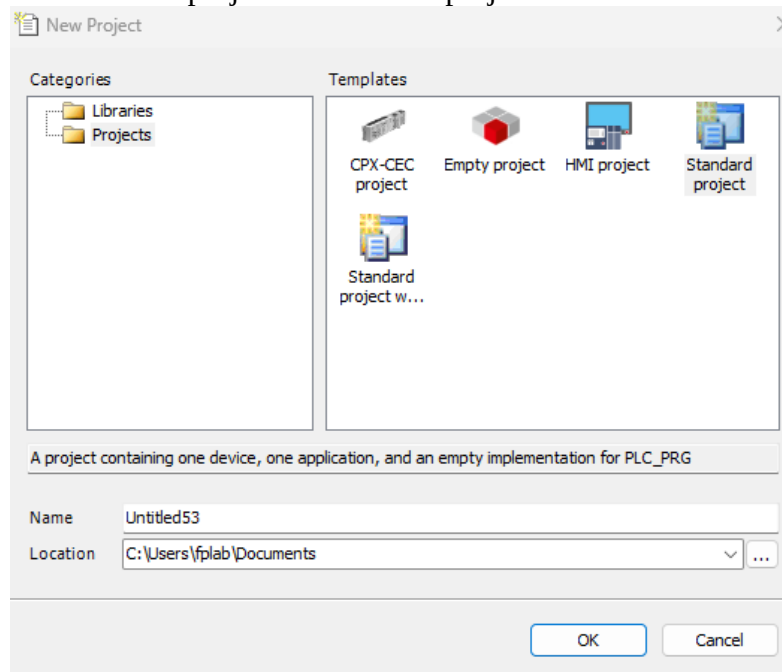
Factory I/O enables the fast construction of virtual 3D factories by selecting among more than eighty configurable parts. Also, Factory I/O comes with over twenty ready-to-use typical industrial applications (also called scenes) to practice real-world control tasks.

Sequence of operation:**STEP-1**

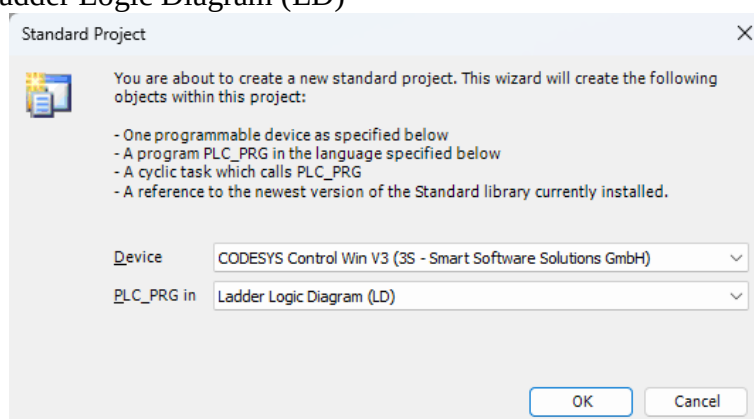
Open codesys software available on the machine.

STEP-2

Click on the file → new project → Standard project.

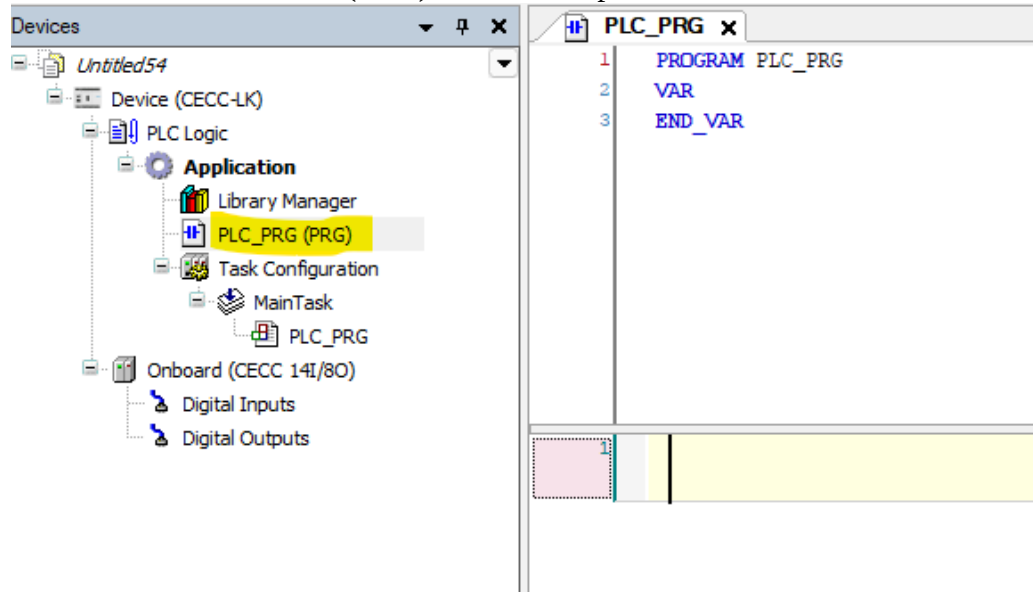
**STEP-3**

Select device as CODESYS Control Win V3 (3S-Smart Software Solutions GmbH)
PLC_PRG in → Ladder Logic Diagram (LD)

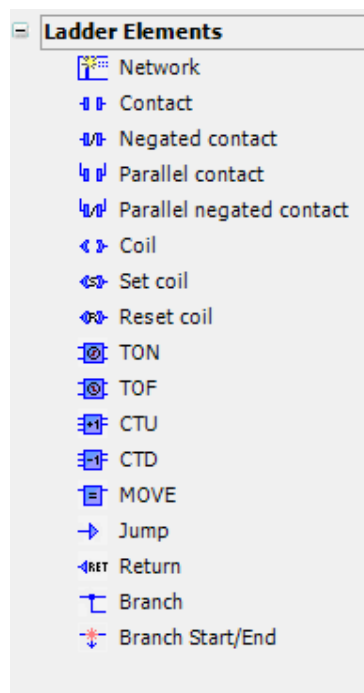


STEP-4

Make a double click on PLC_PRG(PRG) which will open ladder window

**STEP-5**

- To create a given application, use ladder elements such as NO (Normally Open), NC (NormallyClosed), coils, setcoil, and resetcoil.

**STEP-6**

- Provide variable address as shown below

The screenshot shows the 'Auto Declare' dialog box. The 'Scope' is set to 'VAR', 'Name' is 'A', and 'Type' is 'BOOL'. The 'Object' is 'PLC_PRG [Application]'. The 'Address' field contains '%IX0.0'. The 'Flags' section has three checkboxes: 'CONSTANT', 'RETAIN', and 'PERSISTENT', all of which are unchecked. The 'Comment' field is empty. The 'OK' and 'Cancel' buttons are at the bottom right.

- Complete the plc program as per factory application

STEP-7

Communication setup

- Enable Codesys gateway and start plc from the right corner icon tab
- Click on devices in codesys and scan for devices → select PC as a device
- Right click on add devices select Ethernet adapter → select Ethernet → add ModbusTCP slave

STEP-8

- Click on built → Rebuilt to verify the program
- Go into simulation mode use the online → Simulation
- Again use online → Login
- To debug the code use debug → Start
- Finally verify the Ladder using created switches and lamps in the visualization tab

Result:

“ The Shradha with which we perform our actions outwardly unveils the treasure hidden with us “ - AMMA